Supporting document

Project Scenario:

Clean Dental (CD) has its head office in South Melbourne and 5 Dental Clinics in Victoria, recently they have started to experience frequent network and System failures. Manager reported the issues to the outsourced IT department, but they haven't been able to give a solution immediately. Hence, they have posted an advert in Service Seeking website.

CD Currently have numerous single purpose servers taking up racks in a central control room such as Activity Directory server, Email Server running on ired mails, Microsoft Terminal Server for remote access, and Dental system called Best Practice running on a windows server 2019 and another server running for veeam backups. Employees currently using windows 10 & 11 as the client operating system.

The client is concerned about the increasing costs of running physical servers and server failures. There have been increases in utility power bill, maintenance costs, aging equipment, and continual hardware failure that is hurting the company's bottom line. Clean Dental (CD) has asked you to upgrade their legacy hardware server and network to something that is better suited to their future expansion. Currently they have got five Dell PowerEdge T130 physical servers to run all the services.

In addition to the hardware issues, company also notices that the Terminal Server performs really slow for remote users and required to increase the resource allocation when migrate to virtual environment and a firewall in OVF file need to be deployed to protect the network company. CD would like to move the physical servers to virtual servers to prevent the failures and running cost. Also, it will give more flexibility to manage the server infrastructure.

The client would like you to bench test the correct solution for their network and present your finding in report that will make recommendations that can be implemented.

Part A: Documentation.

1. Identify the customer requirement, company structure from the above case study. Your teacher will act as the representative of CD to get more details.
2. Analyse the existing server infrastructure and document it.
3. Based on the above analysis and customer requirement prepare a feasibility report to implement virtual server and network.
4. Prepare and review the compatibility of services on virtual infrastructure.
5. Identify two different virtualisation techniques to implement the solution. List the product and equipment specification to implement it.
6. Propose the preferred platform with a brief reason to get client approval.
7. Identify the technical requirement for the selected virtualisation platform including High Availability.
8. Document the environmental requirement to install and configure virtualisation solution.
9. Design and confirm the virtual network and servers for the customer requirement.
10. Plan, design, and document for high availability failover configuration.
11. Plan, design, and document appropriate backup solution.
12. Prepare a site access request for environmental analyses.
13. Prepare a detailed task list to carry out the project including plan, design, and Implementation stages.
14. Prepare detail task plan with timeline for implementation.
15. Document the installation and provide user documentation.
16. Prepare a document for client sign – off and approval with recommendations and site access notification.
17. Create two test plans to confirm the implementation align with customer request.
18. Analyse current customer infrastructure and determine level of protection, vulnerability, risk, and mitigation according to customer requirements.
19. Develop and document critical infrastructure protection plan, then submit protection plan to required personnel for approval.

Introduction, Overview, and analysis

Clean Dental is a dental company with 5 clinics across Victoria.

The company has its head office in South Melbourne which has a central control room with a physical single purpose servers taking up racks.

The current structure is running on 5 Dell PowerEdge T130 physical servers as below:

- Active Directory server on Windows Server 2019
- Email Server running on ired mails.
- Microsoft Terminal server for remote access
- Best Practice Dental system running on Windows Server 2019
- Veeam backup running on Windows Server 2019
- Employees using Windows 10&11 as the client operating system.

The current structure recently started to have frequent network and system failures which could not be fix with an immediate solution by the IT department. This issue increasing costs of running physical server while having failures and hurting the company's bottom line. Also, they experience slow performance for remote users over the Terminal Server due to low resources, hence they need a solution with increase resources when migrate to virtual environment. Finally, they need a firewall in OVF to protect their network.

Some of the impacts of those issues are:

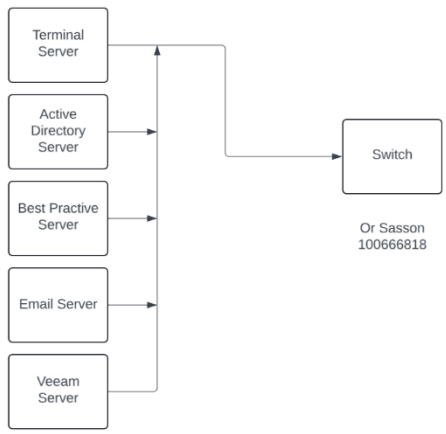
- Utility power bill
- Maintenance costs
- Aging equipment
- Slow performance which impacting the business operations.
- Scalability issues
- Low in resources
- More downtime

The company wish to have a modern structure by moving their physical servers to a virtual server which can give better flexibility and scalability when managing the server infrastructure to prevent failures and running costs.

By implementing a modern solution, the business will experience an improvement of the following:

- Cost will be decreased.
- Better flexibility when managing the server infrastructure.
- Scalability options
- Less downtime and better performance

The current infrastructure with running on 5 Dell PowerEdge T130 physical server:



Compatibility and Feasibility report

The current infrastructure of the physical server needs to be considered when changing it to a virtual server to make sure a smooth migration between the old to the new one and to avoid failure and loss of data.

The company is using 5 physical servers of Dell PowerEdge T130 which supports up to 64GB DDR4 ECC memory which can ensure the smooth operation and handle multitasking. The hardware could be upgraded and have more RAM (up to 64GB) which seems to be enough for the current and the future needs of the company. Upgrading the current server will lower the cost of the project. The storage can up to 16T of storage with all four hard drives installed which should satisfy current and future needs.

Clean Dental seems to use supported operating systems as Windows Server 2019 and Windows 10 & 11 as the client for the employees. Those operating systems are still supported by Microsoft and get regular updates and patches and can be migrated easily to the new virtual server infrastructure. Windows Server 2019 could be upgraded to the latest Windows Server 2022 without any compatibility issues and the Windows 10 & 11 could remain as they are and interact with the new sever infrastructure. In regard of the backup software “Veeam”, It could be implemented on the virtual server as well without any major issues as well as the ired email server.

The same network could be used and modified for the virtual environment. It is possible to keep the current network and IP address on the new server infrastructure.

After my investigation and the information before me I can confirm that all five physical servers with its resources could move to the new virtual server infrastructure. However, some modifications need to be implemented to support some changes and new technologies. VMware converter or veeam could be used for moving the old physical server as they are to the virtual environment.

Virtualization Technologies

Microsoft Hyper-V and VMware ESXi are the main hypervisors which can be used for the implementation of virtual servers. In this section I will compare between the two and provide pros and cons for each hypervisor before deciding which one is better to use for your company.

ESXi – pros

- Known for its stability and reliability when dealing with critical workloads.
- Good performance and optimization, as it has features like memory overcommitment, CPU hot add and storage thin provisioning for improving workload performance.
- Offers scalability which allow to add and remove virtual machines easily.
- Centralizing the management tools in one place which makes it easy for administrators to monitor and configure the environment from a single interface.
- Built-in high availability features such as vSphere HA, which can minimize downtime due to failure in one host.
- Has the VMware Site Recovery Manager which provide disaster recovery for ESXi by automating the failover and recovery to make sure the business can keep operating.
- VMware got a large range of products, tools, and third-party systems which can integrate with ESXi and provide additional functionality.

ESXi – cons

- ESXi is expansive, especially for large organizations which needs to have more advanced features which can higher the cost.
- The ESXi can be heavy and require a lot of resources even after the performance optimization. CPU and memory overhead could occur for hypervisor operations and management.
- Training must be provided, especially for administrator without a lot of experience with hypervisor platforms to be able to manage and troubleshoot ESXi environment.
- Vendor Lock-in as it can be challenging and costly to move to alternative platforms.
- There are some hardware compatibility issues as ESXi does not support all hardware components. It is important to purchase the right hardware according to VMware's Hardware Compatibility List (HCL) before the deployment of ESXi.
- The complexity of ESXi can make challenging when it comes to managing and configuring the environment, especially for large-scale deployments which require advanced features and products.

Microsoft Hyper-V – pros

- Can integrate perfectly with Windows operating systems, which could make it easier to deploy and manage for organizations which already using Windows Server and Microsoft products.
- Cost-effective: Because Hyper-V is a feature part of Windows Server editions, the licenses might be cheaper for organizations already using Windows server.
- It is easy to add and remove VM's at any time which gives a great scalability when using Hyper-V.
- High Availability features such as Failover Clustering can minimize the downtime of critical workload due to failure.
- Hyper-V Replica allowing the organization to replicate VM's to a different site for disaster recovery purposes which can minimize the risk of losing data.
- Hyper-V proves great performance optimization tools such as Dynamic Memory, which can adjust the memory allocation over VM's based on demand for instance.
- Hyper-V supports hyper-converged infrastructure deployment, which can combine all resources into a single platform which can make it easier for administrator to manage and scale the systems at any time.

Microsoft Hyper-V – cons

- While it has a great support with Windows OS and products, it has a limited support for non-Windows OS and products.
- It can be complex to manage and deploy if administrators do not have enough experience with virtualization.
- Not all hardware components are supported which can make compatibility issues if not following the Hyper-V Hardware Compatibility List (HCL) before purchasing new hardware.
- Its third-party ecosystem not as good as other hypervisor platform such as VMware vSphere.
- As any other hypervisor, Hyper-V can have resource overhead, including CPU and memory overhead.
- Licensing can be complex and costly when choosing other Microsoft products and services, such as System Centre Virtual Machine Manager for centralized management.
- Vendor Lock-in as the organization will rely on Microsoft ecosystem and technologies which can make it challenging and costly if deciding to migrate away from Hyper-V to different platforms.

After reviewing and analysing both virtualization technologies I decided that using ESXi would be better for Clean Dental. The reasons of my recommendation are:

- Even though Clean Dental using Windows OS for the server and the client which can integrate better with the Microsoft Hyper-V deployment, I believe that it would be a better idea to use a platform like ESXi which offers wider range of operating systems and applications including Linux and Windows. This will give the organization more flexibility in case they need to use or move to different OS rather than Windows.
- ESXi considered to be the market leader in the industry with a very good record of reliability and performance.
- ESXi can be more costly than Hyper-V when using advanced features, but after reviewing the requirement of Clean Dental it seems that they don’t need a lot of resources and advanced features which can increase the cost dramatically.
- ESXi known for its wide range of features and advanced features which can give better solutions to the organization in case of needing them in the future.
- ESXi got a great performance and scalability abilities which can be beneficial for the organization for future use, especially if needs higher performance or large-scale deployment.
- ESXi got a good vendor support services with training programs and certification.

Technical requirements

1. Hardware: It is critical to make sure purchasing hardware based on the VMware's Hardware Compatibility List along with sufficient CPU, memory and storage resources to ensure smooth operation of VMs and HA features. Also, for redundancy, redundant power supplies and network interfaces which can improve reliability.
2. Licensing: make sure to purchase the correct licensing based on the features and capabilities needed according to this document.
3. Storage infrastructure: To have shared storage infrastructure such as SAN for VM storage which support VMware vSphere VMFS or VVols for shared storage access.
4. Networking infrastructure: high-speed network infrastructure with the capabilities of fault tolerance and redundancy with supported features like VLAN's, NIC teaming and network redundancy protocols.
5. Virtualization software: VMware ESXi hypervisor installed and configures on the server hardware with setting and management interfaces for centralized administration.
6. High Availability: Have multiple ESXi hosts to allow VMware vSphere cluster and enable the high availability feature within the cluster to make sure VMs automatically restart and take over in a case of host failures or high demand.
7. Shared storage: Ensure all hosts in the cluster can access the sma VMFS or VVols syorage for VM failover.
8. Networking setup: configuring the network in a way which ensure network redundancy and isolation to ensure reliable communication between hosts and VMs. Also, configure vSphere Distributed Switches for managing and controlling the network from one place.
9. Monitoring and management tools: Using VMware vCenter server for centralizing management of ESXi hosts and vSphere cluster. Monitoring health, performance and availability could help administrators to manage and troubleshoot issues.
10. Backup and Disaster recovery solutions: Create a server with Veaam software for regular backup and recovery to protect VMs from loss of data due to failure or cyber incidents.
11. Security management: configure firewall, security policies and access control to protect the virtualized infrastructure.
12. Documentation: document all steps of implementation and configurations as a reference for the IT team in case needed for future use.

For full technical requirements and capabilities, it is recommended to visit the link below:

https://www.vmware.com/resources/compatibility/search.php

environmental requirement to install and configure virtualisation solution

Part of the planning of virtualization solution is to ensure that the environment supports all necessary physical, electrical and operational requirements for a smooth installations and operations.

- The physical space and layout need to include enough space in the server room for the physical infrastructure, including proper racks which can meet all safety regulations and ensure airflow. Also, cooling equipment must be in this room.
- The physical server needs to have a good temperature control to maintain the temperature between 18-27C, along with humidity control to keep the room with humidity level between 40-60%. Also, airflow management must be implemented to optimize cooling efficiency with hot and cold aisle containment. Cooling system must have good capacity which can handle the heat increase from the servers.
- The power supply needs to be reliable with the right capacity to handle all hardware. Better to use a redundant power supply for critical servers. Also, it is recommended to install UPS (Uninterruptible Power Supply) systems to make sure that the servers are protected during power outages. Furthermore, to avoid overloading circuits it is recommended to use PDUs (Power Distribution Units) to distribute power efficiently withing racks.
- Ensuring sufficient network bandwidth to make sure that it can handle increased data traffic from VMs. It is recommended to upgrade it if necessary. Also, implement redundant network connections which can ensure high availability and failover management. The cabling must be high quality such as Cat6 or higher to ensure a good networking connection.
- Access control must be in place so only authorized people could access this room with key card access or biometric scanners for instance. It is also required to have surveillance cameras to know what is going on in the room at any time. Fire suppression systems must be in place as well as environmental monitoring which can provide details regularly about the temperature, humidity and air quality.
- All local regulations such as local building codes, electrical standards and safety regulations along with industry standards with the guidelines for data centre design and operation muse by followed.
- The server room should be designed in a way that can be scalable for future growth with enough space, power and cooling capacity for an easy upgrade.
- Back up recovery solution must be in place to be able to protect the data and the infrastructure after any disaster which could happen. Disaster recovery plan should be in place.

Virtual Network and Server Design for Clean Dental

- Network needs to have segmentation with management network, production network, backup network to ensure backup traffic for best performance and security, remote access network for users to connect to the terminal server and VLANs implementation to improve security and reduce the broadcast traffic.
- The VMs will be connected to the virtual network by virtual switches and the management of the network will be centralized across all hosts with VMware Distributed Switches service.
- Firewall will be deployed in the OVF format and network access control will be in place.
- For current needs, we will use 32 to 40GB of RAM and up to 12T of storage. For future needs it could be increased up to 64GB of RAM and 16T of storage.

The server design will be as the following:

Server	VMware ESXi 7 server with Vcenter
Memory	40GB
Storage	8T
Network Adapter	NAT
Ip Address	192.168.15.60

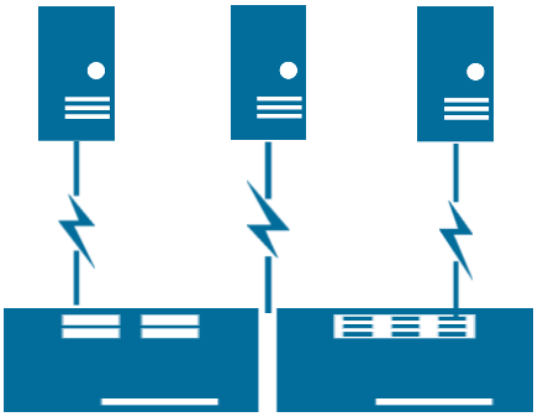
Server	Host 2
Memory	40GB
Storage	8T
Network Adapter	NAT
Ip Address	192.168.15.61

Server	Host 3
Memory	16GB
Storage	8T
Network Adapter	NAT
Ip Address	192.168.15.62

Server	Vcenter SAN server with Veeam backup
Memory	32GB
Storage	12T
Network Adapter	NAT
Ip Address	192.168.15.80

High availability design

As you can see in the diagram below, there is a storage connected to all three severs to make sure that if one of the servers is disconnected due to a failure, the other servers will still be able to access the storage without any interruptions. Every virtual machine on the server which failed will migrate automatically to the other server and operate normally.



Backup solution

For backup purposes, we will be using Veeam software with the following plan:

Data	Source	Software	Type	Destination	Frequency	Retention
Windows 10/11	Host03 with all VMs	Veeam	Full	VeeamBackup hard disk	Weekly	1 years
Email server	Host01 with all VMs and vCenter	Veeam	Full	VeeamBackup hard disk	Daily	Forever
Terminal server	Host02 with all VMs	Veeam	Full	VeeamBackup hard disk	Daily	Forever
Storage and files	vCenterSAN	Veeam	Full	VeeamBackup hard disk	Weekly	Forever

Site access request

As part of the process of upgrading the server infrastructure od Clean Dental, it is required to perform an environmental analysis to ensure that the new infrastructure is supported. To do so we will need to access the server room in the South Melbourne head office.

During the analysis, we will assess the current infrastructure and conditions which might affect the new server installations, along with power and cooling requirements. We will also check that every new installation will meet regulations and industry standards to make sure the environment is safe.

This analysis should take day or two and will be performed during business hours.
Clean Dental team will escort the analysis team during their visit.

Please confirm the receiving of this request and let us know which time is suitable for you so we can arrange this analysis at your earliest convenience.

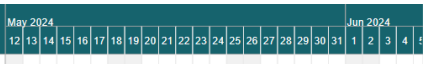
For any questions, please contact Or Sasson.

Task List

- Purchase the server's new hardware components.
- Purchase the correct ESXi licensing.
- Make sure the server room meet the requirements (temperature etc…)
- Install new components such as memory and hard drives.
- Connect the server to the switch.
- Install the ESXi servers.
- Configure network ports and services to run ESXi server remotely.
- Connect and configure the ESXi using vSphere web client and vSphere application.
- Configure the virtual network, test it and document the outcome.
- Create and install virtual machine on the ESXi servers.
- Patch and update the installed operating systems.
- Add all servers to the vSphere.
- Test the virtual server and connect the network connectivity between the server and to the internet.
- Create a snapshot of the terminal server.
- Convert the RDS server to a virtual server.
- Create the virtual firewall.
- Create and configure High Availability
- Test the High Availability is working as expected.
- Perform live migration of the email server between the host to make sure it works.
- Configure and test the backup server using Veeam software.
- Implement, test and document the protection plan.
- Enable ESXi Shell (DCUI) and limit access via enforcing strict access security policies.

Task plan with timeline

	Task	Assigned To	Start	End	Dur
	Implementation	Or	5/15/24	5/28/24	9.5
1	Purchase the server hardware	Or	5/15/24	5/15/24	0.5
2	Purchase the correct ESXi licensing	Or	5/16/24	5/16/24	
3	Make sure the server room meet the requirements	Or	5/16/24	5/16/24	1
4	Install components such as memory and hard drives	Or	5/16/24	5/16/24	1
5	Connect the servers to the switch	Or	5/17/24	5/17/24	1
6	Install the ESXi servers	Or	5/18/24	5/18/24	
7	Configure network ports and services to run ESXi server remotely	Or	5/18/24	5/18/24	
8	Connect and configure the ESXi using vSphere web client and vSphere application	Or	5/18/24	5/18/24	
9	Configure the virtual network, test it and document the outcome	Or	5/18/24	5/18/24	
10	Create and install virtual machine on the ESXi servers	Or	5/19/24	5/19/24	
11	Patch and update the installed operating systems	Or	5/19/24	5/19/24	
12	Add all servers to the vSphere	Or	5/20/24	5/20/24	1
13	Test the virtual server and connect the network connectivity between the server and to the internet	Or	5/20/24	5/20/24	1
14	Create a snapshot of the terminal server	Or	5/21/24	5/21/24	
15	Convert the RDS server to a virtual server	Or	5/21/24	5/22/24	2
16	Create the virtual firewall	Or	5/22/24	5/22/24	1
17	Create and configure High Availability	Or	5/23/24	5/23/24	1
18	Test the High Availability is working as expected	Or	5/23/24	5/23/24	1
19	Perform live migration of the email server between the host to make sure it works	Or	5/24/24	5/24/24	
20	Configure and test the backup server using Veeam software	Or	5/25/24	5/25/24	
21	Implement, test and document the protection plan	Or	5/26/24	5/27/24	1
22	Enable ESXi Shell (DCUI) and limit access via enforcing strict access security policies	Or	5/28/24	5/28/24	1



User documentation

Please refer to part B

Client Sign-Off and Approval Document

Project: Upgrading and moving physical servers to virtual servers
Date: 25/05/2024
Client: Clean Dental
Project manager: Or Sasson

This document is a formal request for sign-off and approval of the recommendations and planning of this project which was requested by Clean Dental to move the physical server to virtual servers. This sign-off request includes the site access for environmental analysis as was described above.

By signing off this document, you allow the project manager, Or Sasson, to proceed with the project after accessing the server room in the head office for environmental analysis.

Please complete this request withing 10 business days, otherwise we will assume you are no willing to proceed and no further action will be required by you.

For any concerns, please contact the project manager, Or Sasson.

Approved by:

Business name: Clean Dental
Name:
Date:
Signature:

First test plan

Host/VM	Description	Expected results
VMware ESXi 7 server with Vcenter	Pinging between all internal servers and hosts and pinging to external websites such as Google	Successful pings internally and externally
Host02		
Host03		
Vcenter SAN + Veeam backup server		
VM01, VM02 and VM03	Pinging between all VMs and between the VMs to the hosts and to the internet	Successful pings internally and externally

Second test plan

Service/features	Description	Expected results
Live migration	Perform a live migration from host 1 to host 2	The VM should migrate to host 2 while it still operating without any disruptions
High Availability	High Availability failover by shutting down one of the hosts with VMs	VMs from the failed host should move to alternative host which is up
snapshoot	Create a snapshot of one of the VMs	Successful creation of the snapshot
Backup and restore with Veeam	Backup one of the VMs with after creating file, then delete the files and restore the VM with the backup files	Successful backup and restore with the original files which were created before backup

Analysis of Clean Dental's current infrastructure and the level of protection, vulnerability, risk and mitigation

The current structure is running on 5 Dell PowerEdge T130 physical servers as below:

- Active Directory server on Windows Server 2019
- Email Server running on ired mails.
- Microsoft Terminal server for remote access
- Best Practice Dental system running on Windows Server 2019
- Veeam backup running on Windows Server 2019
- Employees using Windows 10&11 as the client operating system.

Network security was not specified, and it require an upgrade including the deployment of a new firewall in OVF format.

Currently, Clean Dental has experienced network and system failures with the current infrastructure including very slow remote access. This analysis will analyse the level of protection and identify vulnerabilities and risks including mitigation strategies related to the current infrastructure.

Protection level:

The servers are hosted in a central control room, but due to the aging hardware there are maintenance issues which suggest physical protection is compromised due to frequent failures. Also, due to the hardware failures, the Veeam backup solution might won't be reliable enough for a good level of data protection. The network security also missing protection, as they need a new firewall deployment.

The current infrastructure got the following risks and vulnerabilities:

- The Dell PowerEdge T130 is aging which causes downtime and high maintenance costs.
- User productivity is affected as the terminal server for remote access is too slow.
- Weak network security measures are in place (such as firewall) which increase the risk of data breaches and unauthorized access.
- The power and cooling systems might not work well and overheat the equipment which can cause aging.
- The data management such as data retention period may not comply with industry standards and regulations.

To mitigate those risks and vulnerabilities, it is recommended to move all services to a virtualized environment with hypervisor such as VMware vSphere to improve resource allocation, to be less dependent on the hardware and to increase scalability and reliability. Also, allocating additional resources such as CPU and RAM to the terminal server could improve user's experience and perform better. Furthermore, deploying a firewall from OVF file can protect the network better against cyber threats. It is also important to upgrade the power and cooling systems to optimize it and reduce the risks of hardware overheating. Maintaining and monitoring the systems regularly with the right monitoring tools, could indicate about issues on time and avoid down time. Moreover, it is recommended to review and update data retention policies and to make sure the business comply with best practices so unnecessary sensitive data isn't stored for a long time which can reduce data breaches.

Be aware of all those vulnerabilities and risks and follow the mitigation suggestions, could ensure that the upgrade of the system will improve the security and performance of the systems along with reducing the management costs.

Critical infrastructure protection plan for Clean Dental

In the previous part of this document, we analysed the current risks and vulnerabilities of the current infrastructure and suggested some mitigation solutions. In this part we will outline the protection plan and ask for an approval.
This protection plan will make sure that after the upgrade is done it will be ensured that the availability, integrity and confidentiality of critical system and data along with minimalizing the risks of network and systems failures. Also, better security measures will be in place to avoid unauthorized access and cyber threats and ensuring that the infrastructure will have scalability for potential future growth.

The protection plan applies to servers, network infrastructure, data protection, remote access and environmental controls in the severs room.

This can be done by following those steps:

- Understanding of the critical assets of the business such as servers, network devices and data storage.
- Making sure all tools and software such as firewalls, IDS/IPS and backup are up to date and ready to use.
- Implementing physical security such as access control with keycard access and security cameras.
- Ensuring the control of the environment with cooling, humidity control, and fire suppression systems.
- Configuring the OVF firewall file and segment the network with VLANs for network security.
- Use antivirus software on all servers and workstations. Also, apply patches and updates regularly.
- High availability configured to ensure the availability of all services.
- Encrypt sensitive data on rest and transit and implement Veeam backup solution.
- Monitor and logging the systems regularly with IDS/IPS and log management.
- Always ensure network connectivity internally and externally and test is with ping command.
- Use vulnerability scanning tools such as OpenVAS and perform penetration testing regularly.
- Have incident response plan to mitigate any attack which might occur.

Note that it is important to review the compliance with data protection regulations and best practices regularly and use monitoring tools to be able to indicate any issue as soon as possible to avoid downtime and security incidents.

Please approve this critical infrastructure protection plan so we can proceed with the implementation.

Approved by,

Business name: Clean Dental
Name:
Date:
Signature:

Part B: Implementation

1. Follow the WHS and Environment requirements to Install and Configuring a hypervisor server.
2. Configure network ports and services to run Hyer-V/ESXi management services remotely.
3. Connect and configure ESXi using vSphere web client and vSphere application/Hyper-V using Hyper-V Manger.
4. Configure the virtual network based on the design.
5. Test the virtual network and document the outcome.
6. Create and install virtual machine on the ESXi/Hyper-V server using ISO files.
7. Patch and update the installed operating systems.
8. Test the virtual servers and confirm network connectivity between the servers and to internet.
9. Update the prepared test plans based on the testing outcome.
10. Create snapshot before adjusting the resources on the terminal Server. (You don't need to configure the terminal services-use a fresh virtual server and create snapshot)
11. Monitor and adjust resources according to customer requirement on terminal server to increase user experience.
12. Verify the changes you have made on the virtual machine.
13. Convert the RDS server to a virtual server (This is the convert a physical machine to virtual machine, if you don't have to physical machines, you can simulate using a virtual machine running outside the Esxi environment.)
14. Create the virtual firewall using supplied OVFs into ESXi environment.
15. Enable high availability and show that it is configured and active.
16. Change the VMs IP address and run high availability failover test using the above configured HA.
17. Email server on the first ESXi using more resource, therefor you have decided to migrate it to 2nd ESXi host. Perform a live migration to move the email server from host 1 to host 2. (Again, you don't need to configure email server. Simulate using a virtual machine).
18. Add some files and folders in the newly created virtual machine. Using the backup utility (Eg:Veeam) perform a backup task. Once back up task is finished. Delete some files from the virtual machine. Now restore virtual machine you just backed up to confirm the deleted files can be restored.
19. Run the test plans to ensure the outcomes match with the test plans.
20. Analyse errors and troubleshoot if you have any issues.
21. Implement, test, and document the protection plan
22. Enable ESXi Shell (DCUI) for troubleshooting purposes merely. Limit access via enforcing strict access security policies.

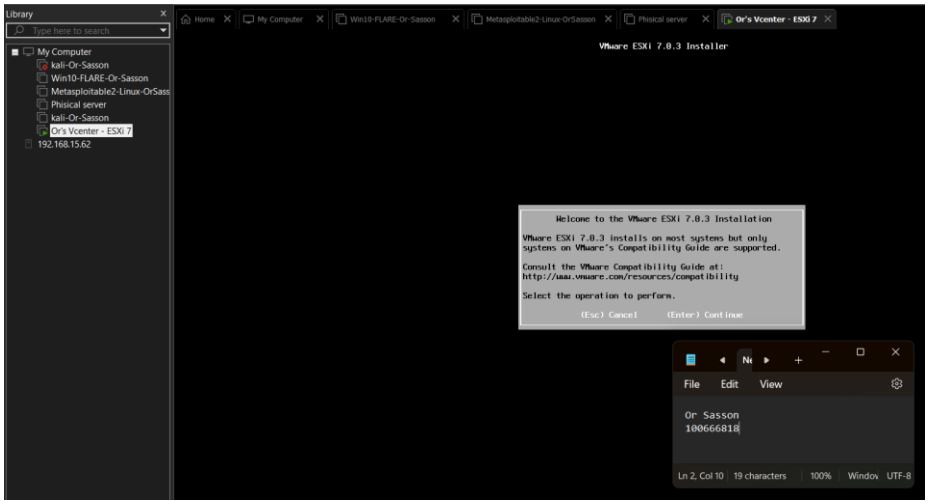
WHS and Environment requirements to Install and Configuring a hypervisor server

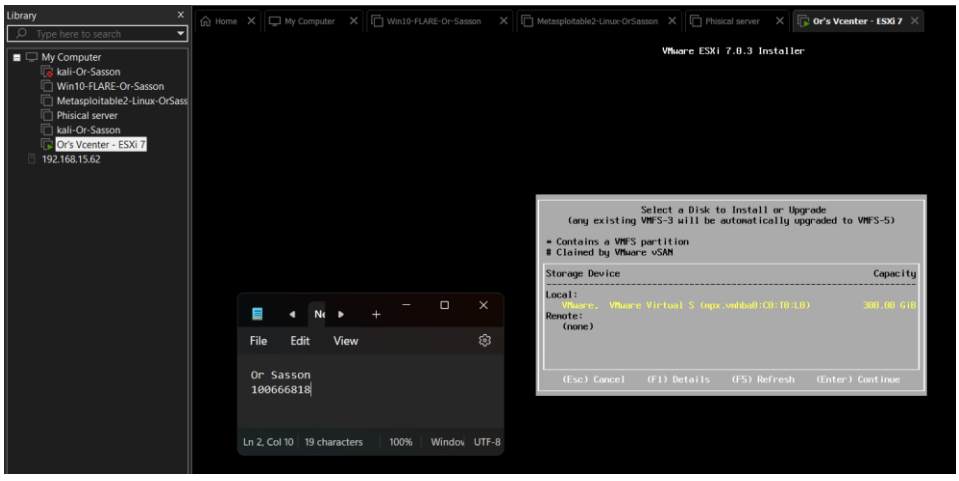
Before configuring the hypervisor server, we made sure that the Workplace Health and Safety (WHS) and environmental requirements are met. While viewing the environment we made sure that all hazards were identified with the process of installation such as electrical risks along with ensuring control measures including lifting techniques, training and using protective equipment.

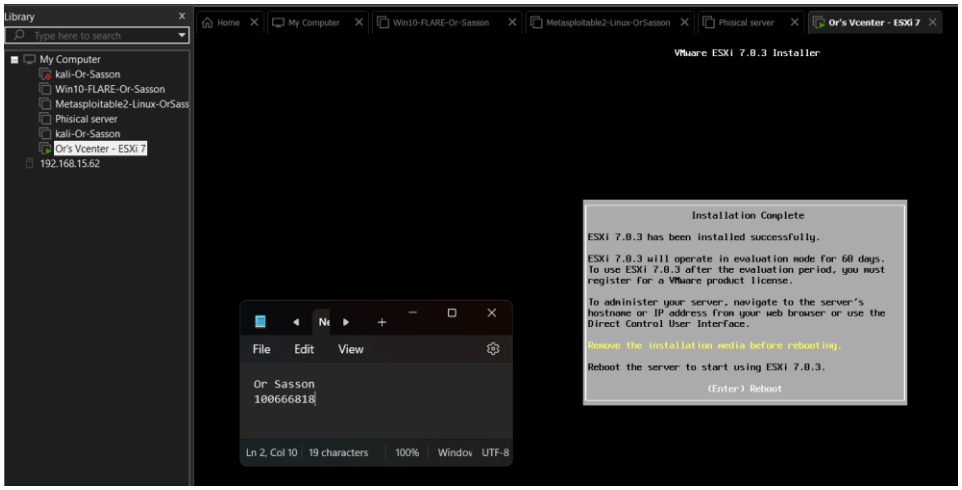
We can confirm that the following requirements are met with the WHS and environmental requirements:

- Electrical safety including isolation procedures, certified electricians, equipment testing.
- Manual handling including lifting techniques and mechanical aids.
- Fire safety including fire suppression systems and fire extinguishers.
- Cable management and workplace design.
- Temperature and humidity control and airflow management to protect the environment.
- Power management including uninterruptible power supply and energy-efficient equipment.
- Environmental monitoring including sensors and alerts.
- Noise control to isolate the room with acoustic insulation and ensure the equipment is quiet.
- Waste management including E-Waste disposal.

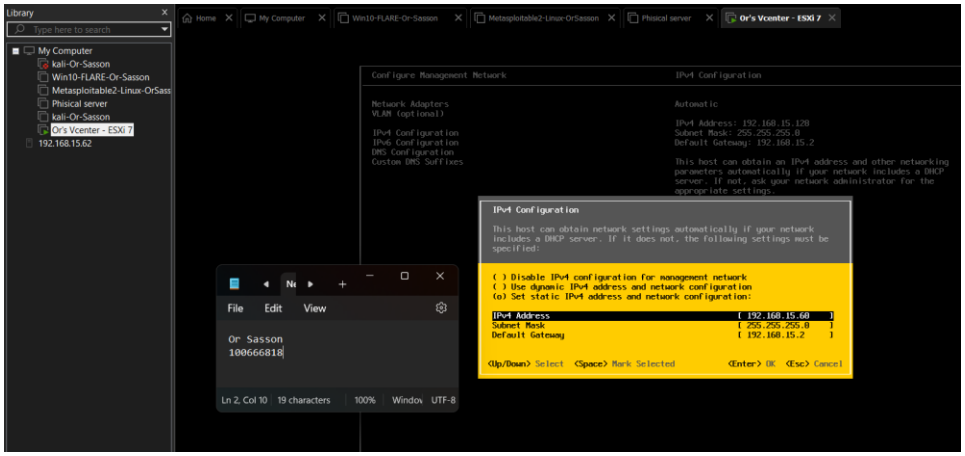
Now it is all set, and we can proceed with the installation process.

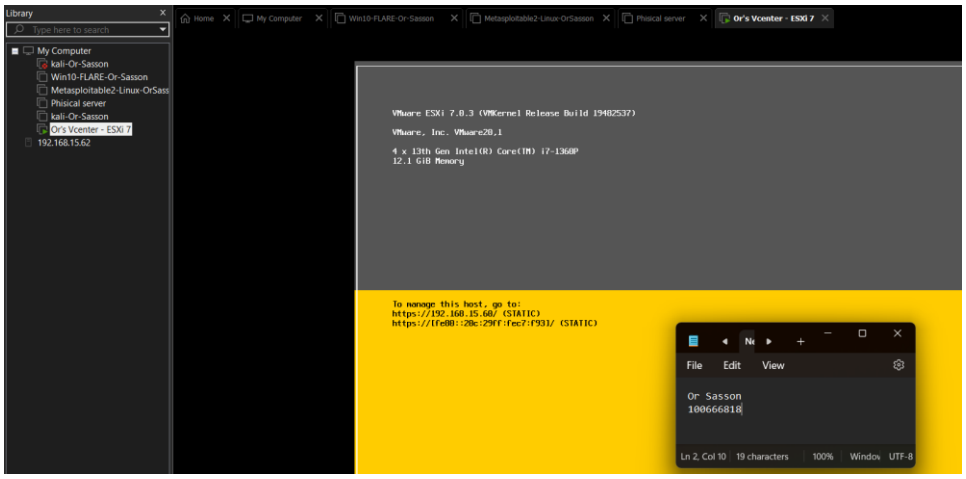


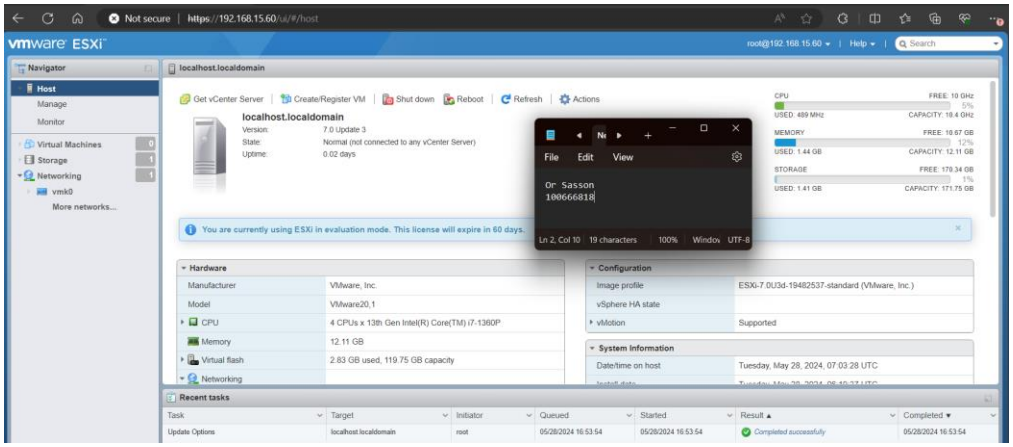



Library
Type here to search
My Computer
kali-Or-Sasson
Win10-FLARE-Or-Sasson
Metasploitable2-Linux-OrSass
Phisical server
kali-Or-Sasson
Or's Vcenter - ESXi 7
192.168.15.62
Home
My Computer
Win10-FLARE-Or-Sasson
Metasploitable2-Linux-OrSasson
Phisical server
Or's Vcenter - ESXi 7
VMware ESXi 7.0.3 Installer
Installation Complete
ESXi 7.0.3 has been installed successfully.
ESXi 7.0.3 will operate in evaluation mode for 60 days.
To use ESXi 7.0.3 after the evaluation period, you must
register for a VMware product license.
To administer your server, navigate to the server's
hostname or IP address from your web browser or use the
Direct Control User Interface.
Remove the installation media before rebooting.
Reboot the server to start using ESXi 7.0.3.
(Enter) Reboot
File
Edit
View
Or Sasson
100666818
Ln 2, Col 10
19 characters
100%
Windov
UTF-8

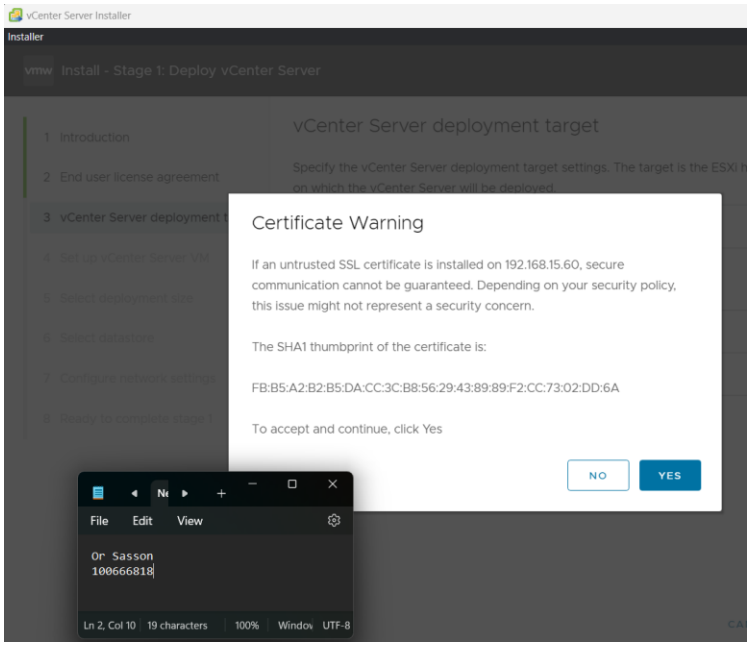
Configure network ports and services to run Hyer-V/ESXi management services remotely.

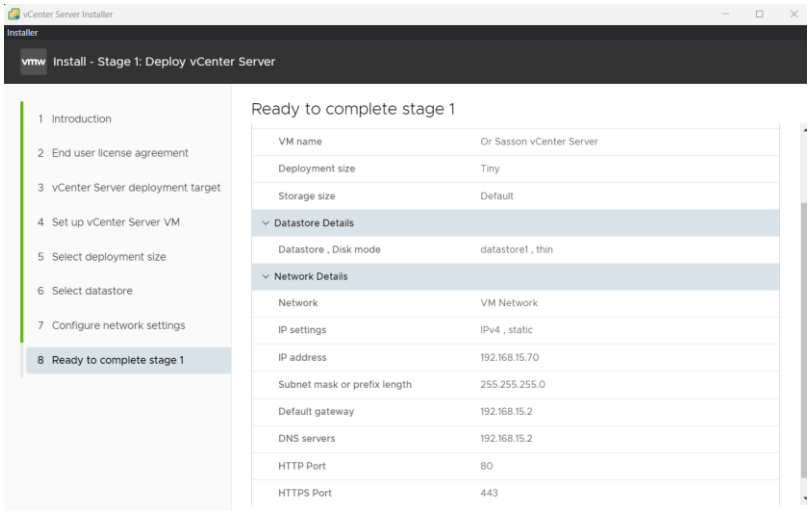


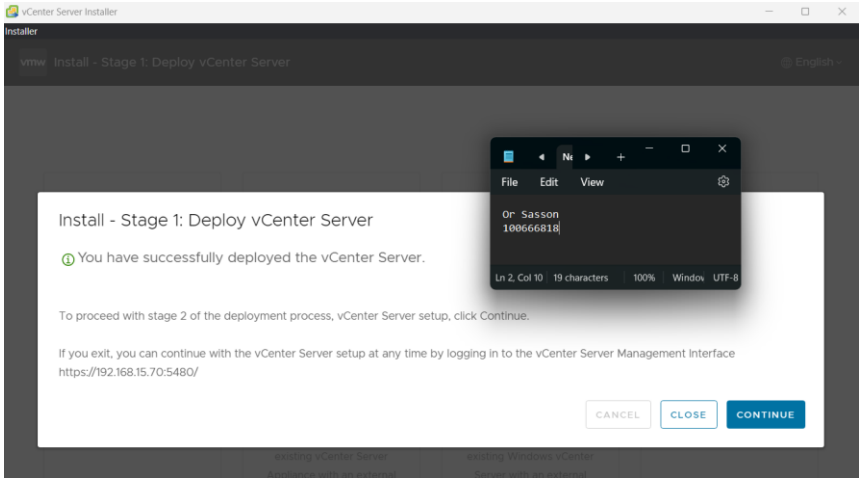


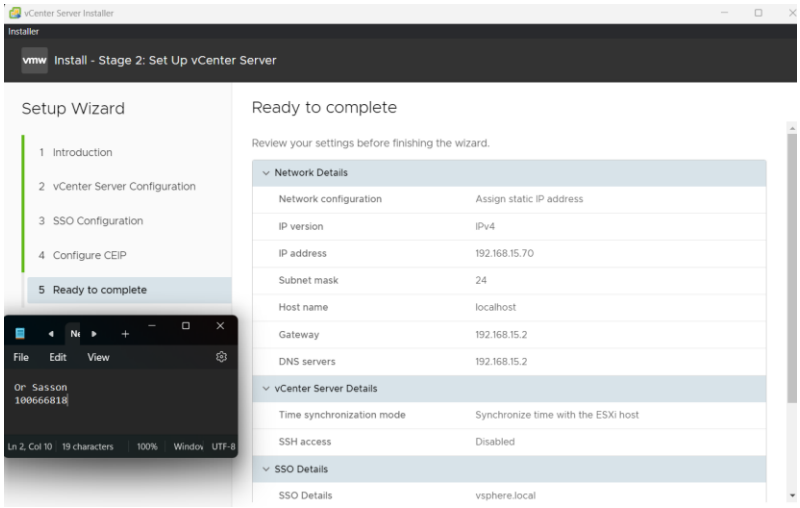


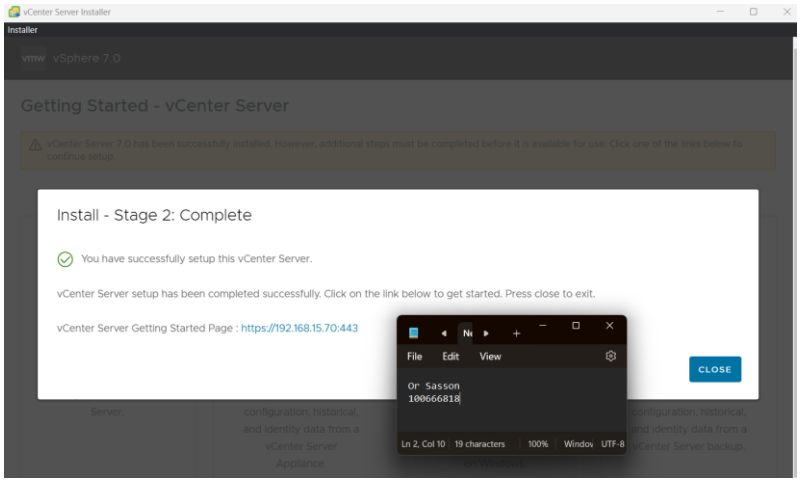
Connect and configure ESXi using vSphere web client and vSphere application/Hyper-V using Hyper-V Manger.

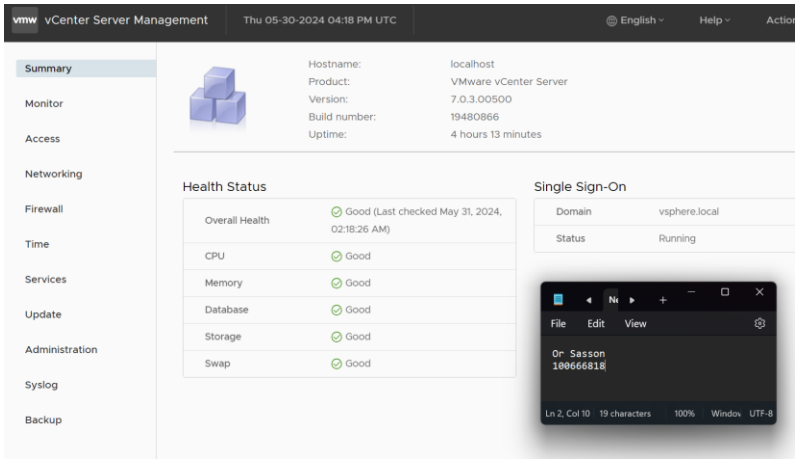




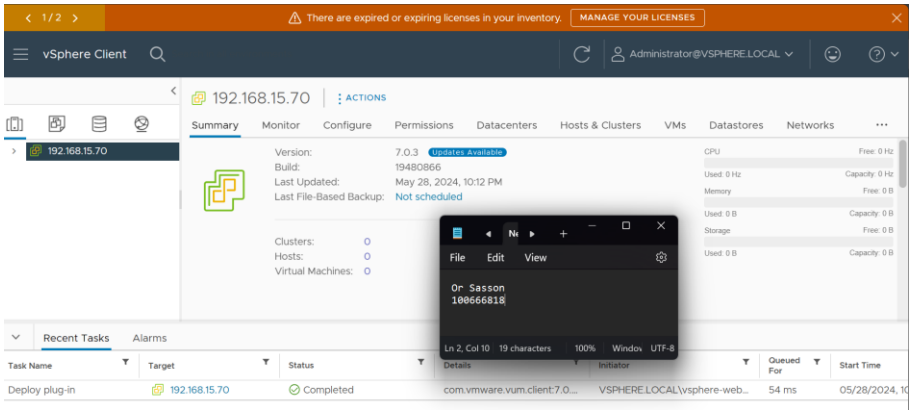


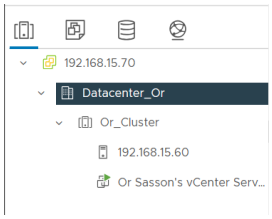

vCenter Server Installer
Installer
Install - Stage 2: Set Up vCenter Server
Setup Wizard
1 Introduction
2 vCenter Server Configuration
3 SSO Configuration
4 Configure CEIP
5 Ready to complete
Ready to complete
Review your settings before finishing the wizard.
Network Details
Network configuration
Assign static IP address
IP version
IPv4
IP address
192.168.15.70
Subnet mask
24
Host name
localhost
Gateway
192.168.15.2
DNS servers
192.168.15.2
vCenter Server Details
Time synchronization mode
Synchronize time with the ESXi host
SSH access
Disabled
SSO Details
SSO Details
vsphere.local
File Edit View
Or Sasson
100666818
Ln 2, Col 10 19 characters 100% Windov UTF-8


vCenter Server Installer
Installer
vSphere 7.0
Getting Started - vCenter Server
Install - Stage 2: Complete
You have successfully setup this vCenter Server.
vCenter Server setup has been completed successfully. Click on the link below to get started. Press close to exit.
vCenter Server Getting Started Page : https://192.168.15.70:443
CLOSE
File Edit View
Or Sasson
100666818
Ln 2, Col 10 19 characters 100% Windov UTF-8

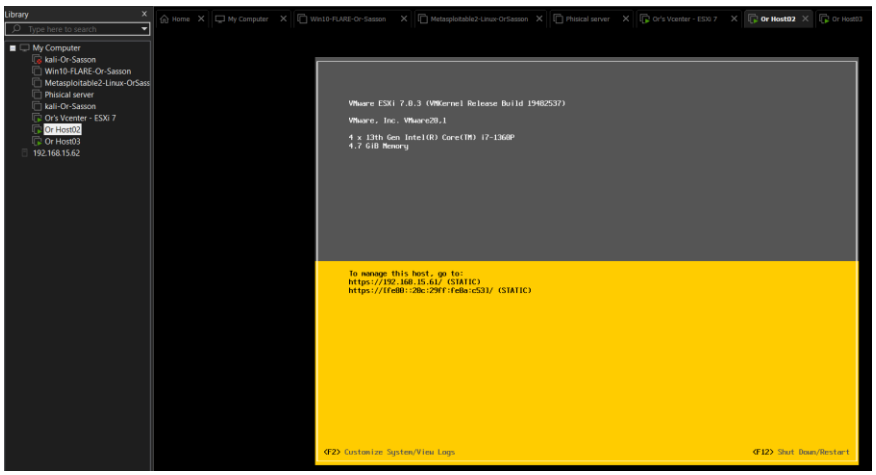

vCenter Server Management
Thu 05-30-2024 04:18 PM UTC
English
Help
Summary
Monitor
Access
Networking
Firewall
Time
Services
Update
Administration
Syslog
Backup
Hostname: localhost
Product: VMware vCenter Server
Version: 7.0.3.00500
Build number: 19480866
Uptime: 4 hours 13 minutes
Health Status
Overall Health
Good (Last checked May 31, 2024, 02:18:26 AM)
CPU
Good
Memory
Good
Database
Good
Storage
Good
Swap
Good
Single Sign-On
Domain
vsphere.local
Status
Running
File Edit View
Or Sasson
100666818
Ln 2, Col 10 19 characters 100% Windov UTF-8

Configure the virtual network based on the design

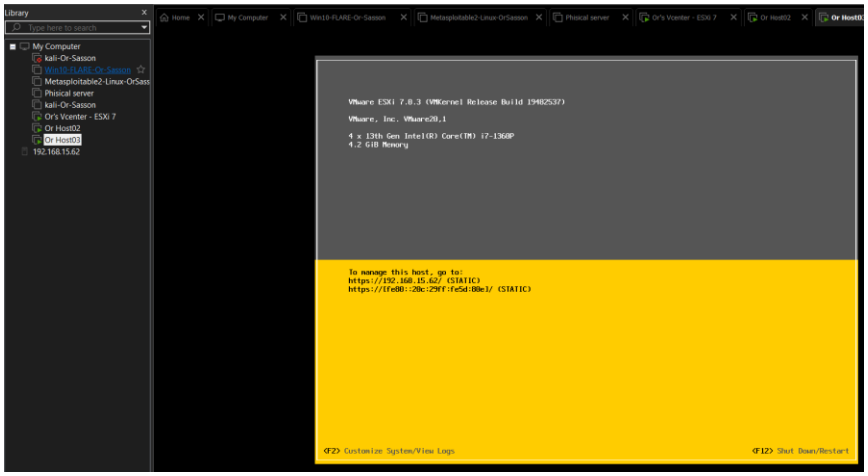


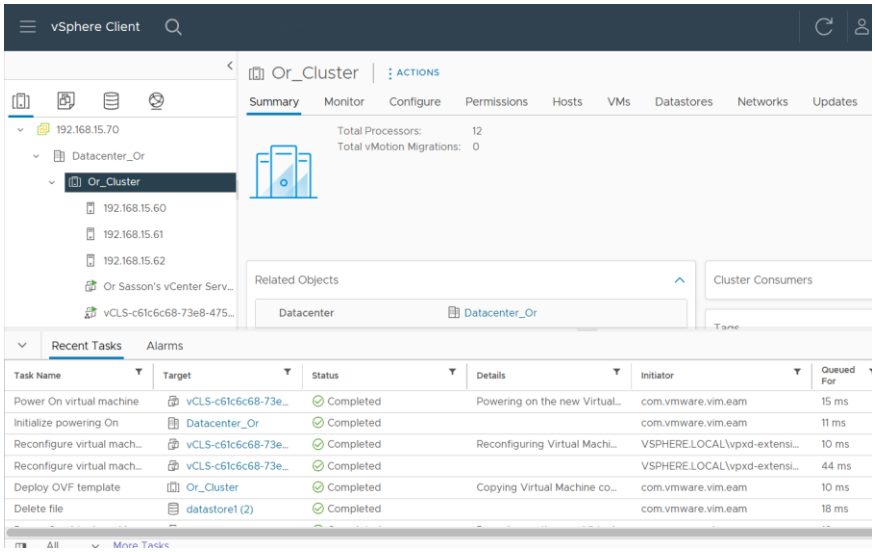


Host02

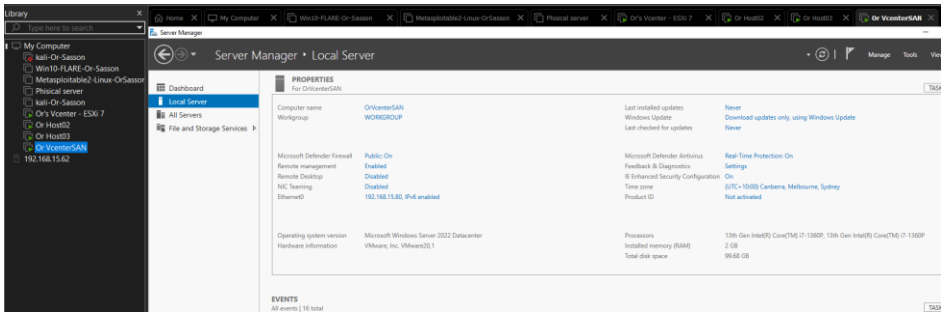


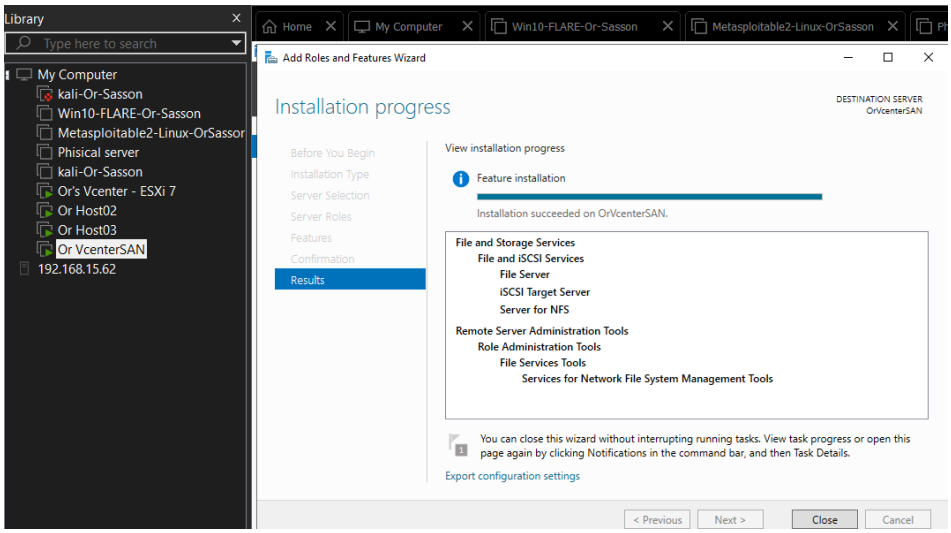
Host03

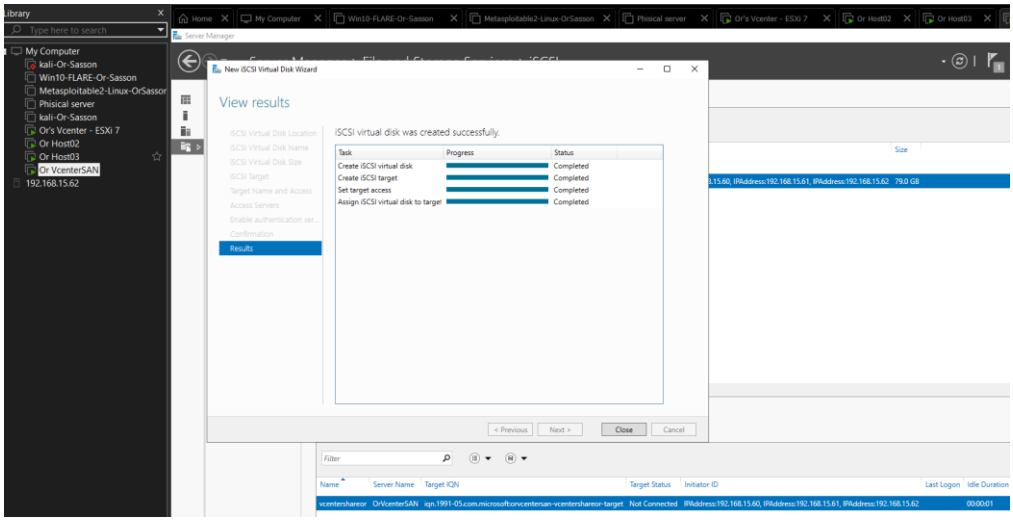



vSphere Client
Or_Cluster
ACTIONS
Summary
Monitor
Configure
Permissions
Hosts
VMs
Datastores
Networks
Updates
192.168.15.70
Datacenter_Or
Or_Cluster
192.168.15.60
192.168.15.61
192.168.15.62
Or Sasson's vCenter Serv...
vCLS-c61c6c68-73e8-475...
Total Processors: 12
Total vMotion Migrations: 0
Related Objects
Datacenter
Datacenter_Or
Cluster Consumers
Recent Tasks
Alarms
Task Name
Target
Status
Details
Initiator
Queued For
Power On virtual machine
vCLS-c61c6c68-73e...
Completed
Powering on the new Virtual...
com.vmware.vim.eam
15 ms
Initialize powering On
Datacenter_Or
Completed
com.vmware.vim.eam
11 ms
Reconfigure virtual mach...
vCLS-c61c6c68-73e...
Completed
Reconfiguring Virtual Machi...
VSPHERE.LOCAL\vpxd-extensi...
10 ms
Reconfigure virtual mach...
vCLS-c61c6c68-73e...
Completed
VSPHERE.LOCAL\vpxd-extensi...
44 ms
Deploy OVF template
Or_Cluster
Completed
Copying Virtual Machine co...
com.vmware.vim.eam
10 ms
Delete file
datastore1 (2)
Completed
com.vmware.vim.eam
18 ms

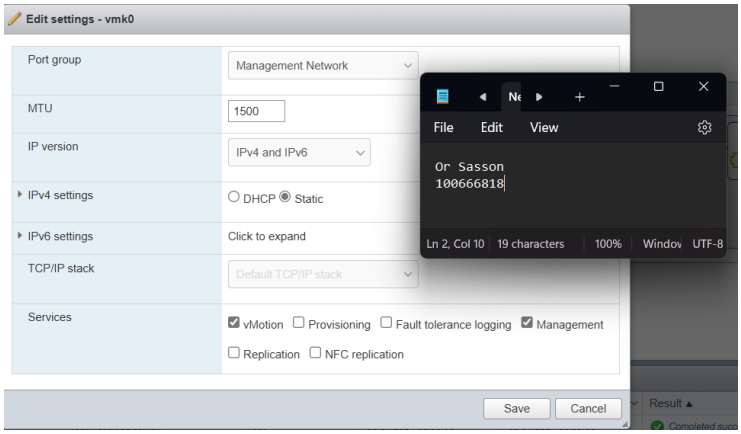
Vcenter SAN


Server Manager • Local Server
Dashboard
Local Server
All Servers
File and Storage Services
PROPERTIES
EVENTS

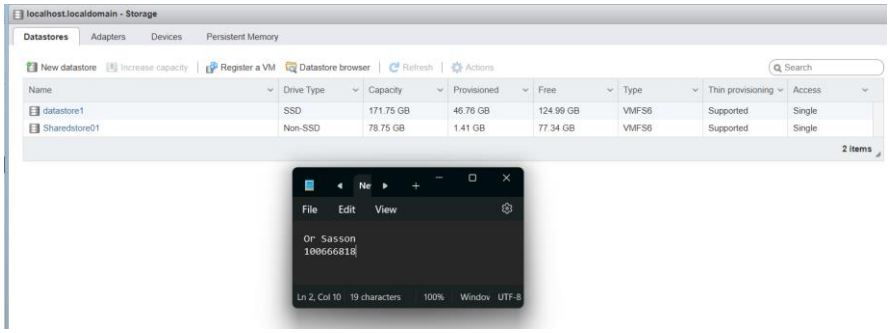

Library
Type here to search
My Computer
kali-Or-Sasson
Win10-FLARE-Or-Sasson
Metasploitable2-Linux-OrSassor
Phisical server
kali-Or-Sasson
Or's Vcenter - ESXi 7
Or Host02
Or Host03
Or VcenterSAN
192.168.15.62
Home
My Computer
Win10-FLARE-Or-Sasson
Metasploitable2-Linux-OrSasson
Add Roles and Features Wizard
Installation progress
DESTINATION SERVER
OrVcenterSAN
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
View installation progress
Feature installation
Installation succeeded on OrVcenterSAN.
File and Storage Services
File and iSCSI Services
File Server
iSCSI Target Server
Server for NFS
Remote Server Administration Tools
Role Administration Tools
File Services Tools
Services for Network File System Management Tools
You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.
Export configuration settings
< Previous
Next >
Close
Cancel



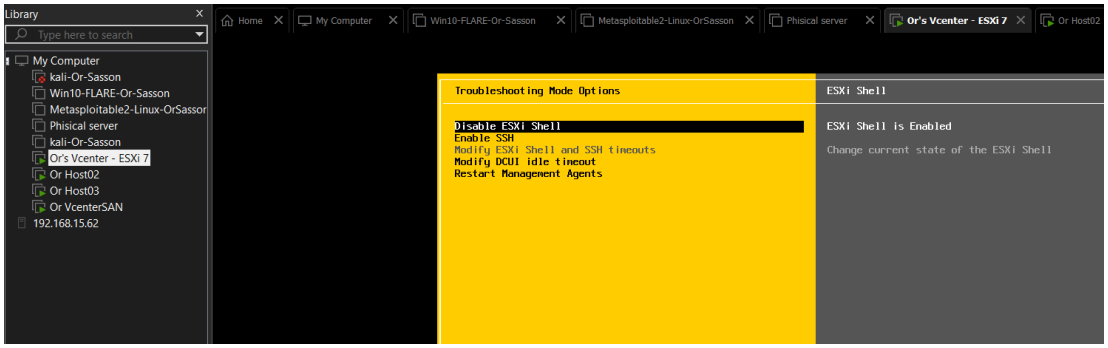
I enabled vMotion on the three hosts.

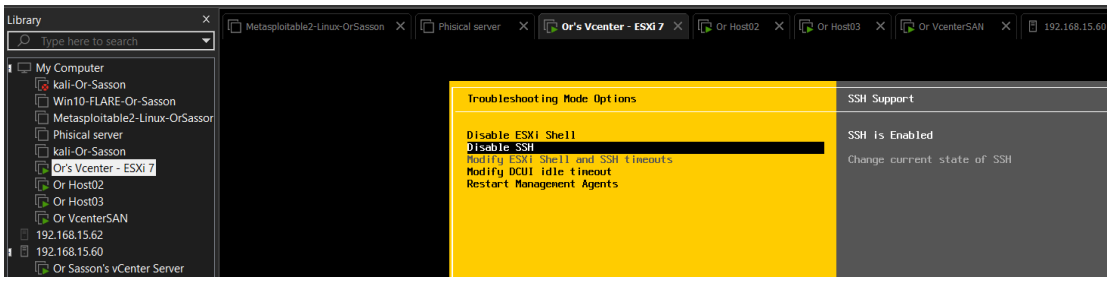


Added the shared storage to all 3 hosts



Enable ESXi Shell (DCUI) for troubleshooting purposes merely and SSH for remote access on all 3 hosts



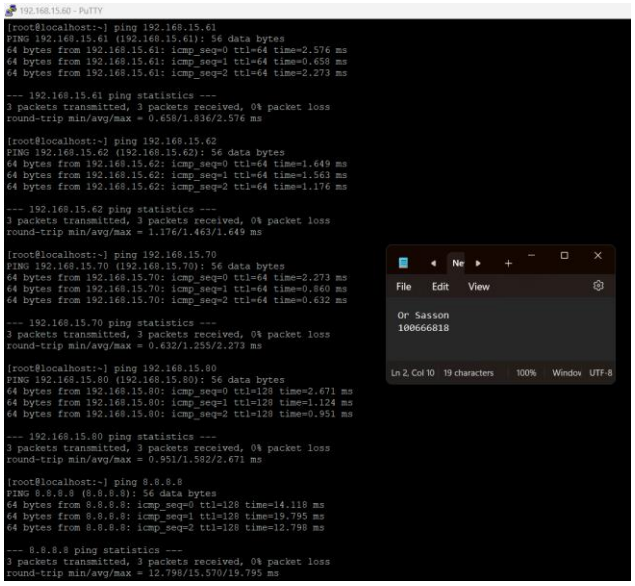


Test the virtual network and document the outcome

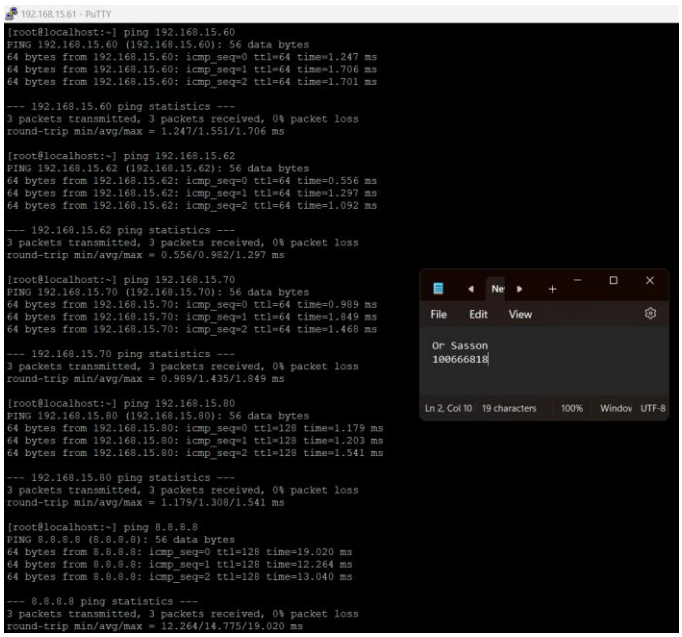
After enabling SSH and Shell, I started SSH connection for each server and testes the pings between all servers and between the servers to the external network. Bellow the following table you will find the screenshots with all results.

Test Type	Description	Expected results	Actual results	Status	Comments\ Issues
Internal ping	Connectivity between the virtual servers	Successful ping responses from all servers	Responded correctly	Successful	No issues.
External ping to 8.8.8.8 (Google)	Connectivity between virtual servers to an external network	Successful ping responses to the external network	Responded correctly	Successful	No issues.

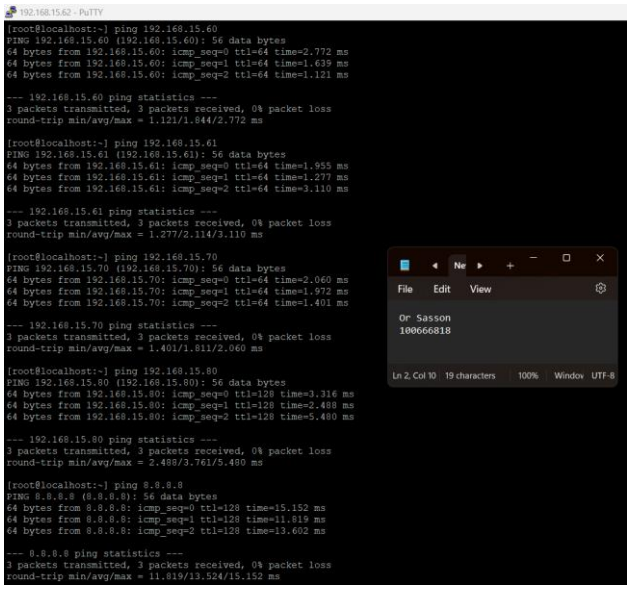
From VMware ESXi 7 server with Vcenter (192.168.15.60):



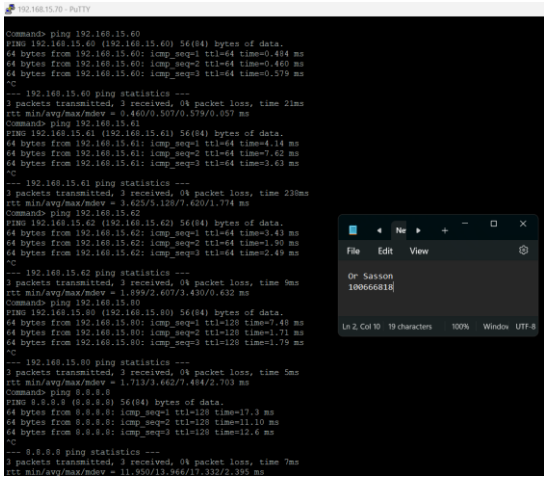
From Host02 (192.168.15.61):



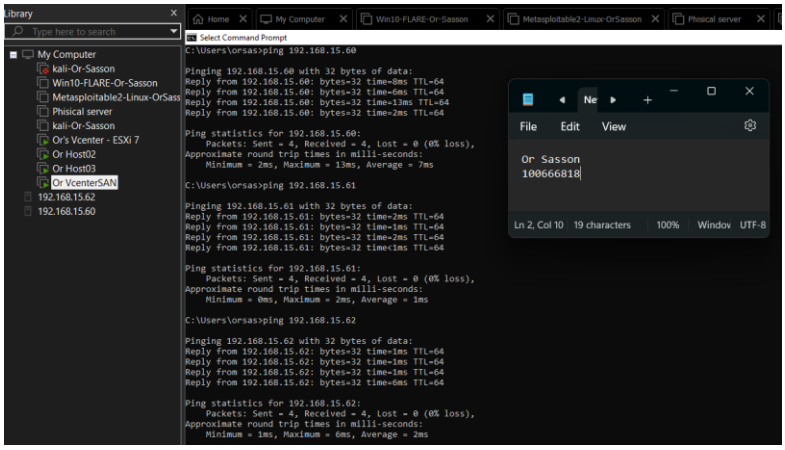
From Host03 (192.168.15.62):

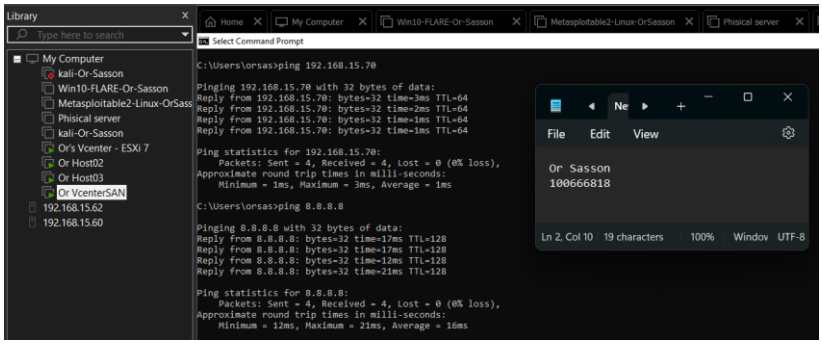


From Vcenter\vSphere (192.168.15.70):



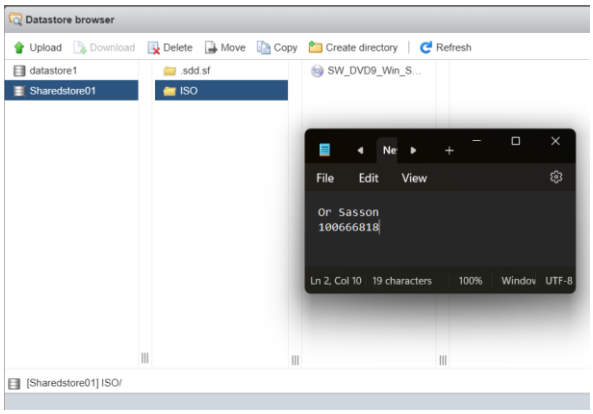
From VcenterSAN (192.168.15.80):



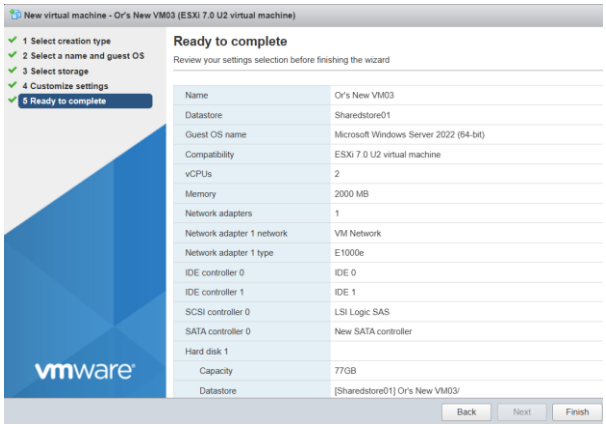

Library
Type here to search
My Computer
kali-Or-Sasson
Win10-FLARE-Or-Sasson
Metasploitable2-Linux-OrSass
Phisical server
kali-Or-Sasson
Or's Vcenter - ESXi 7
Or Host02
Or Host03
Or VcenterSAN
192.168.15.62
192.168.15.60
Home
My Computer
Win10-FLARE-Or-Sasson
Metasploitable2-Linux-OrSasson
Phisical server
Select Command Prompt
C:\Users\orsas>ping 192.168.15.70
Pinging 192.168.15.70 with 32 bytes of data:
Reply from 192.168.15.70: bytes=32 time=3ms TTL=64
Reply from 192.168.15.70: bytes=32 time=2ms TTL=64
Reply from 192.168.15.70: bytes=32 time=1ms TTL=64
Reply from 192.168.15.70: bytes=32 time=1ms TTL=64
Ping statistics for 192.168.15.70:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 1ms
C:\Users\orsas>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=17ms TTL=128
Reply from 8.8.8.8: bytes=32 time=17ms TTL=128
Reply from 8.8.8.8: bytes=32 time=12ms TTL=128
Reply from 8.8.8.8: bytes=32 time=21ms TTL=128
Ping statistics for 8.8.8.8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 12ms, Maximum = 21ms, Average = 16ms
Ne
File
Edit
View
Or Sasson
100666818
Ln 2, Col 10
19 characters
100%
Windov
UTF-8

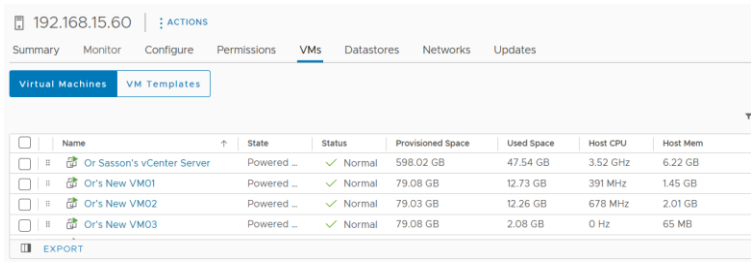
Create and install virtual machine on the ESXi/Hyper-V server using ISO files

First, I uploaded to the shared storage the ISO file of Windows Server 2022



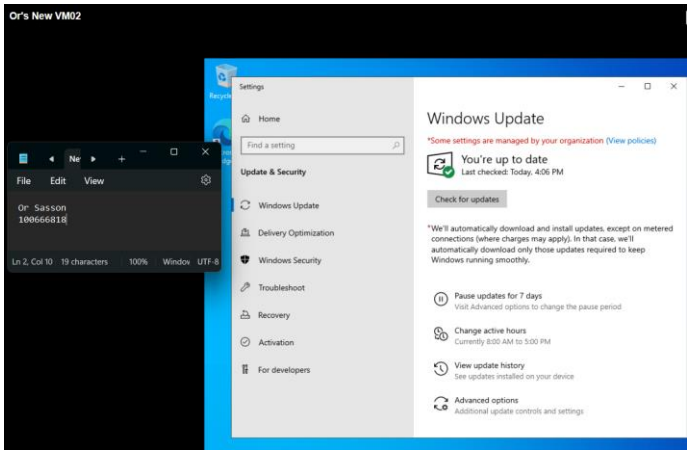
Then, I created 3 VMs on the first host with the vCenter with the Windows Server 2022





VM01 IP: 192.168.15.65
VM02 IP: 192.168.15.66
VM03 IP: 192.168.15.67

Patch and update the installed operating system

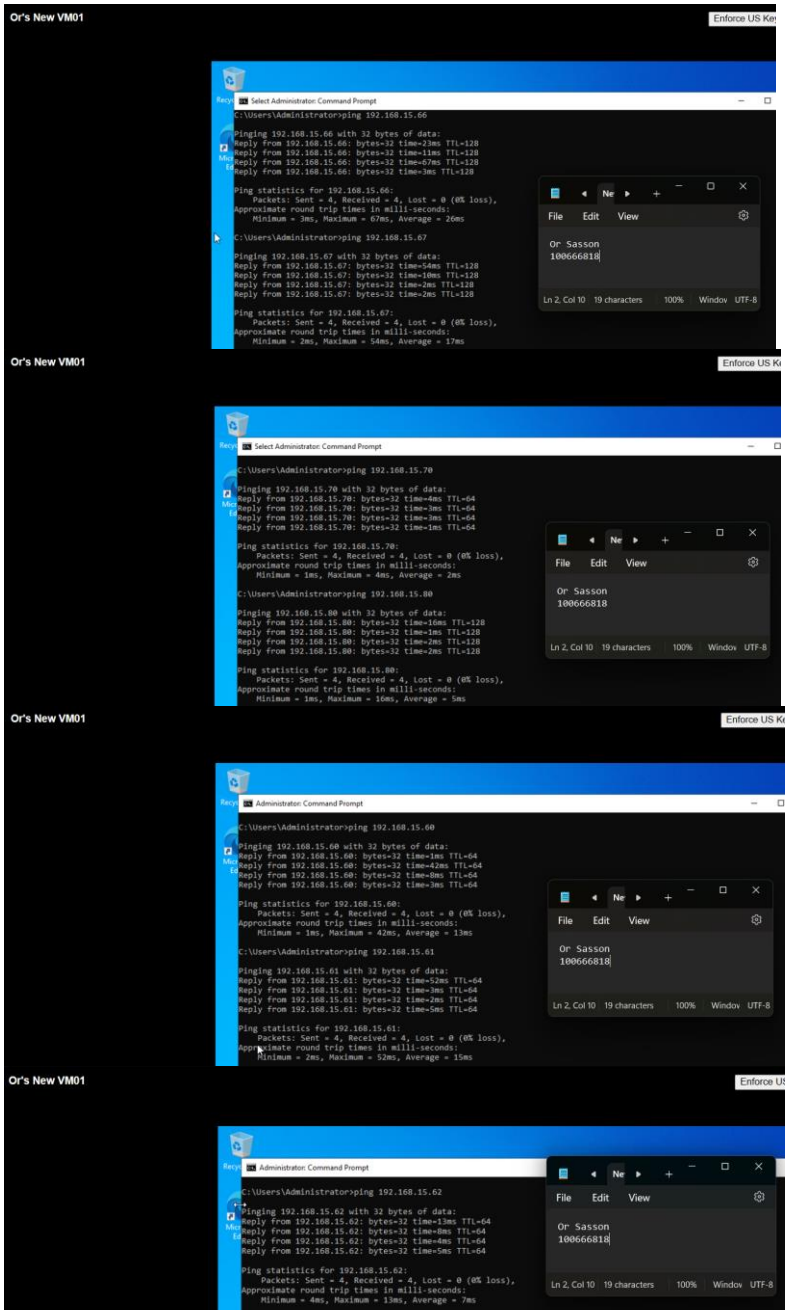


Test the virtual servers and confirm network connectivity between the servers and to internet.

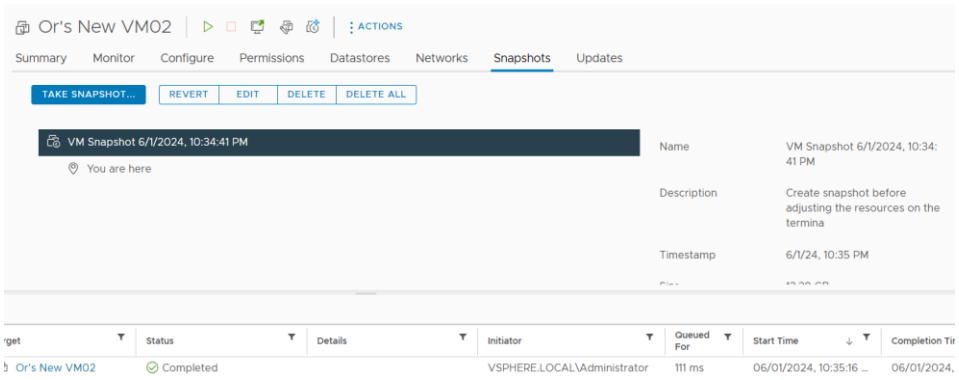
I checked the connectivity between all 3 VMs, between all VMs to the hosts and between all VMs to the internet. Bellow the following table you will find the screenshots with all results.

Test Type	Description	Expected results	Actual results	Status	Comments\ Issues
Internal ping	Connectivity between all VMs and to the hosts	Successful ping responses from all VMs and hosts	Responded correctly	Successful	No issues.
External ping to 8.8.8.8 (Google)	Connectivity between all VMs to the internet	Successful ping responses to the external network	Responded correctly	Successful	No issues.

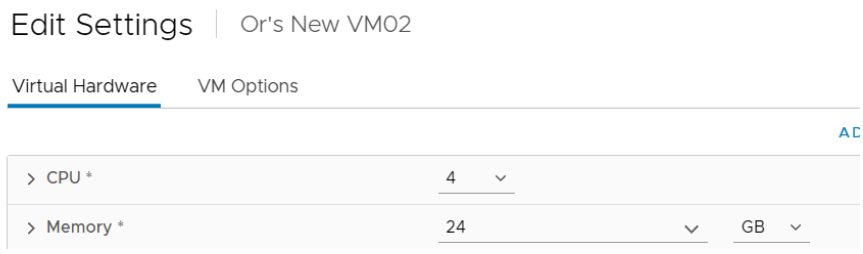
Screenshots of the ping from VM01 to all VMs, hosts and the internet:

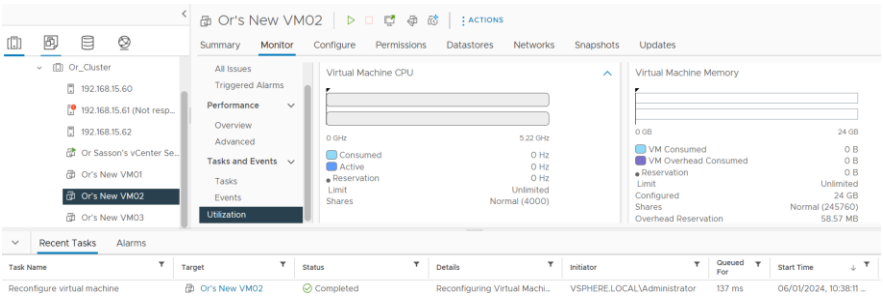


Create snapshot before adjusting the resources on the terminal

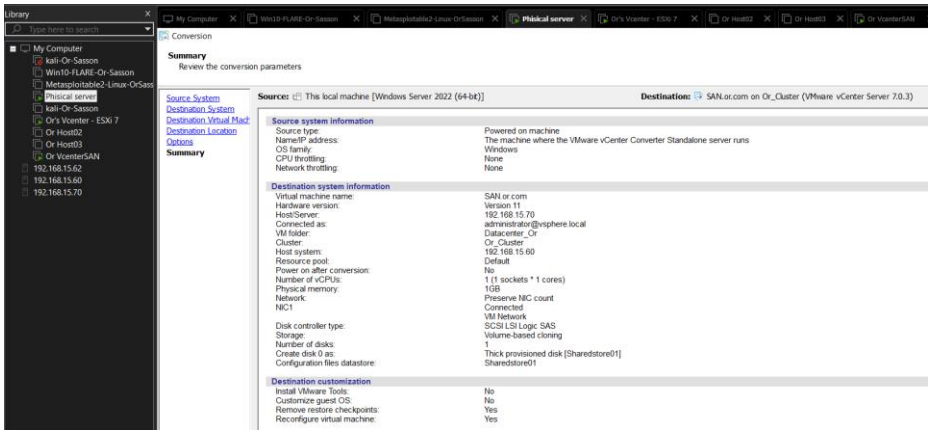


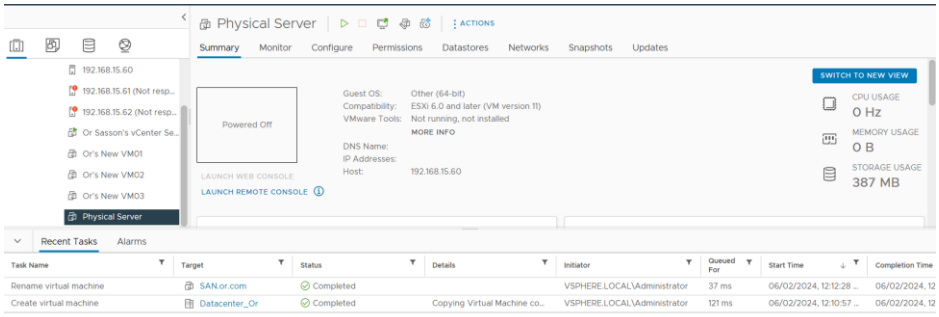
Monitor and adjust resources according to customer requirement on terminal server to increase user experience and verify the changes



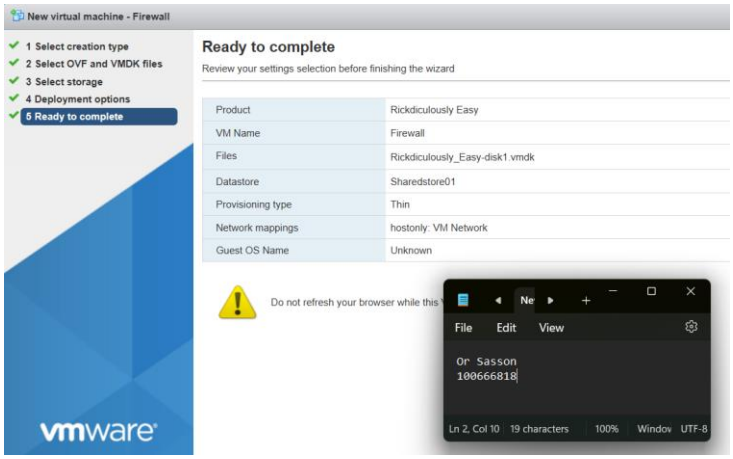


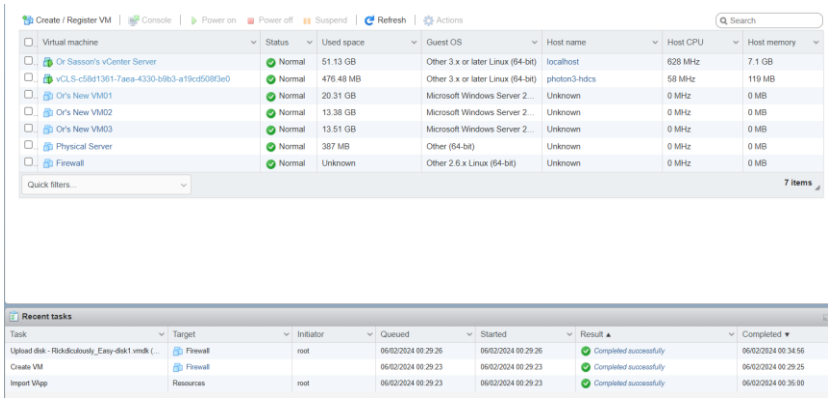
Convert the RDS server to a virtual server



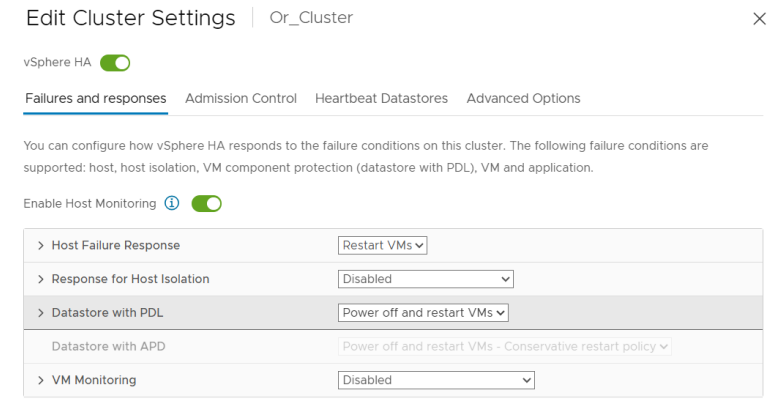


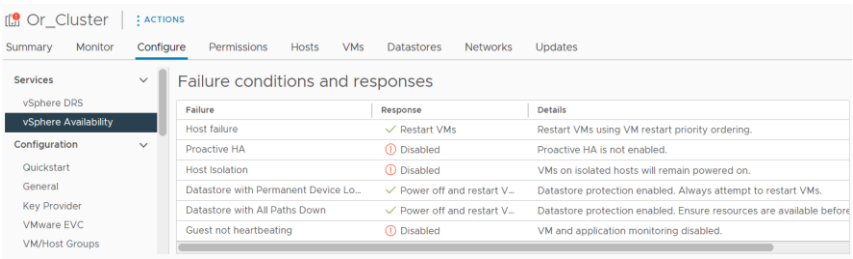
Create the virtual firewall using supplied OVFs into ESXi environment



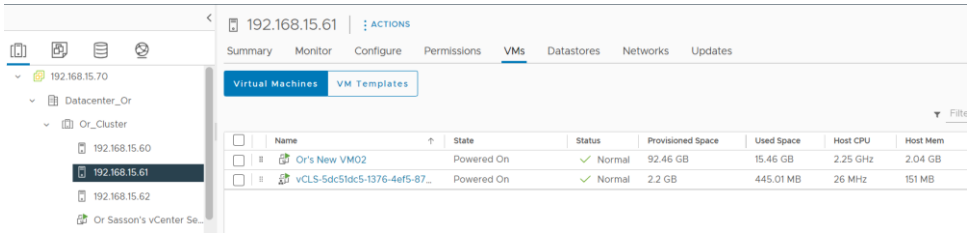


Enable high availability and show that it is configured and active

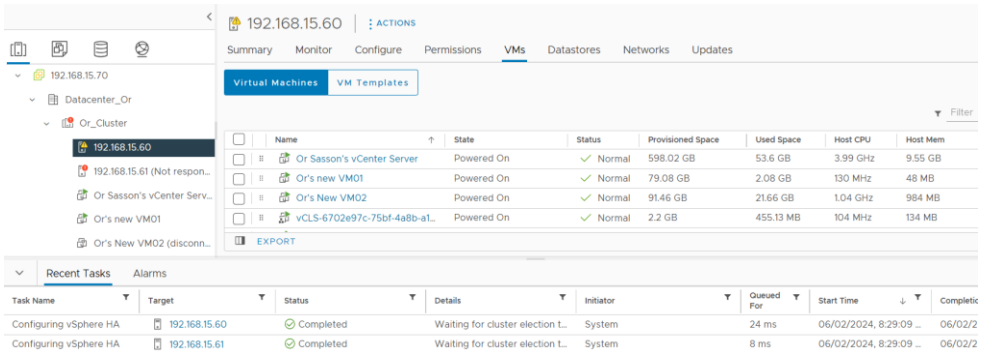




I moved VM02 to the Host02 (192.168.15.61) and powered off host02 to see if the HA is working as expected.

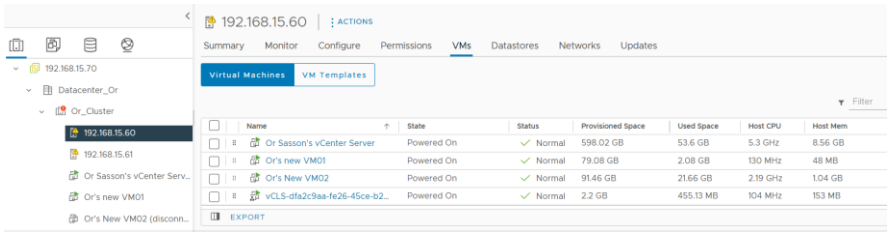


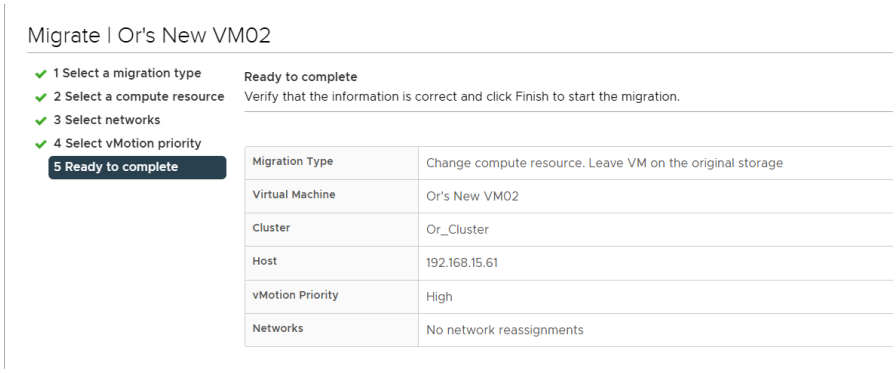
As you can see in the screenshot below, the VM02 moved automatically to host01 (192.168.15.60)

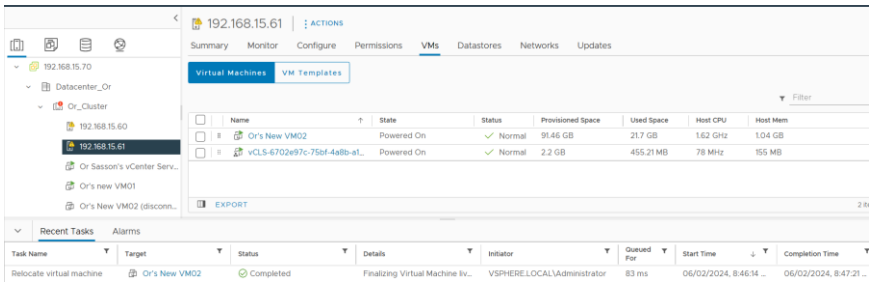


Email server on the first ESXi using more resource, therefor you have decided to migrate it to 2nd ESXi host. Perform a live migration to move the email server from host 1 to host 2.

Live migration of VM02 from host01 (192.168.15.60) to host02 (192.168.15.61):

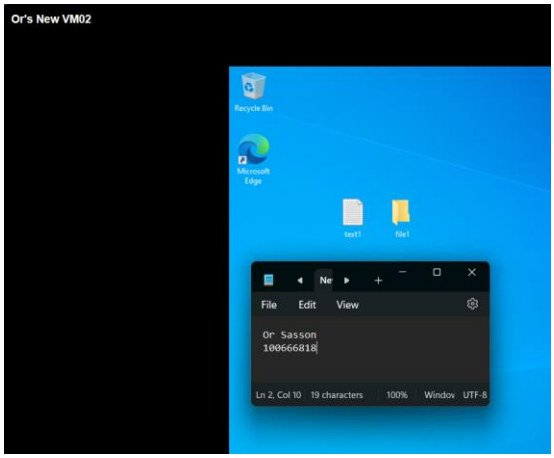




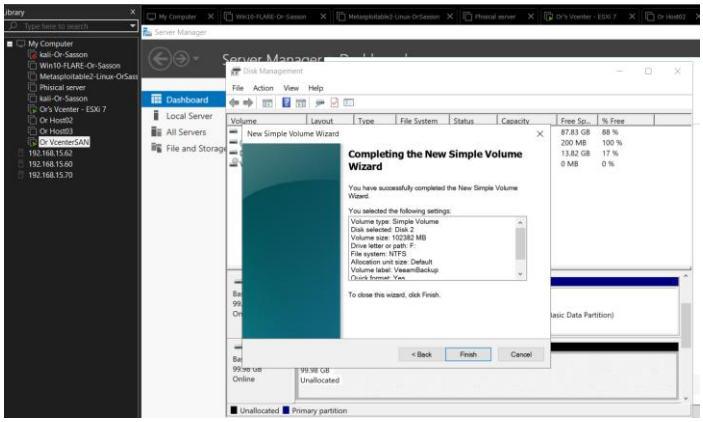


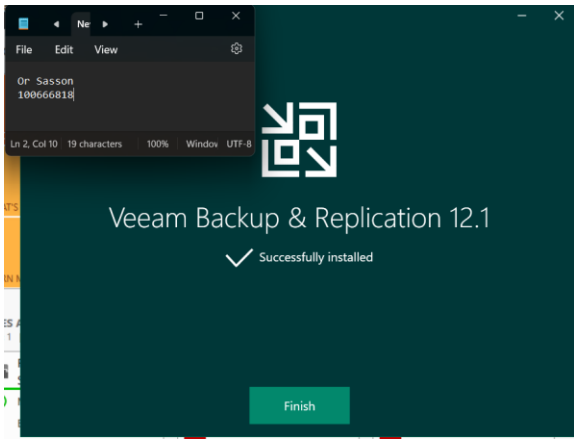
Add some files and folders in the newly created virtual machine. Using the backup utility (Eg:Veeam) perform a backup task. Once back up task is finished. Delete some files from the virtual machine. Now restore virtual machine you just backed up to confirm the deleted files can be restored.

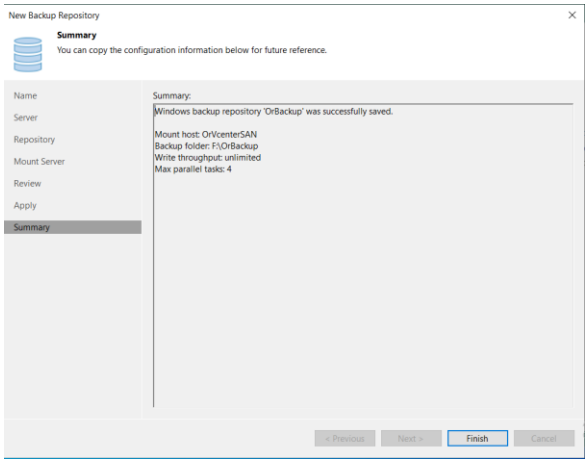
Before the backup, I created a text file and a folder.

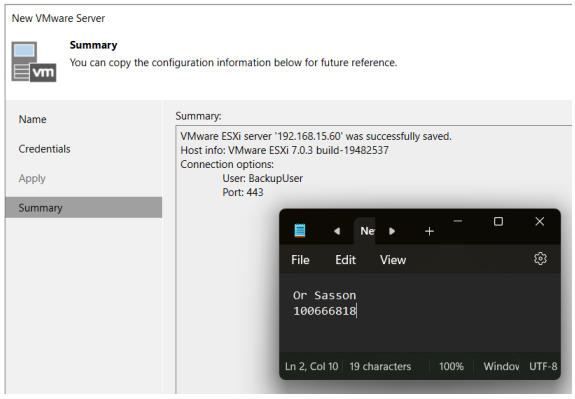


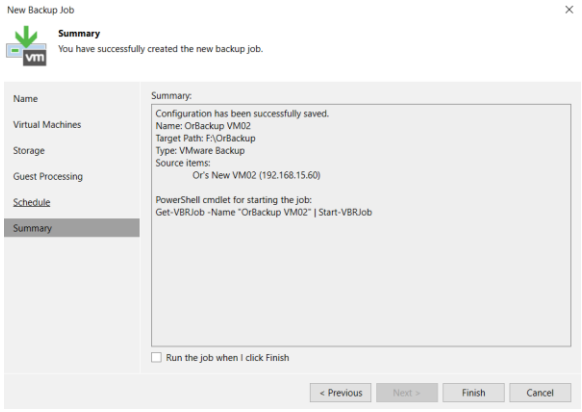
Then, I installed Veeam on the vCenterSAN server.

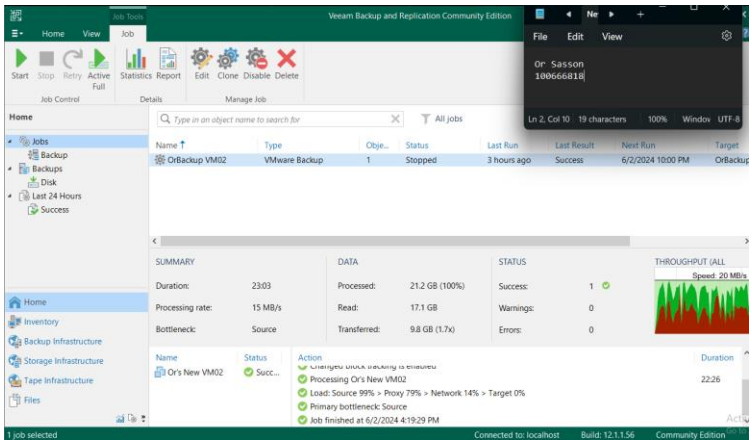




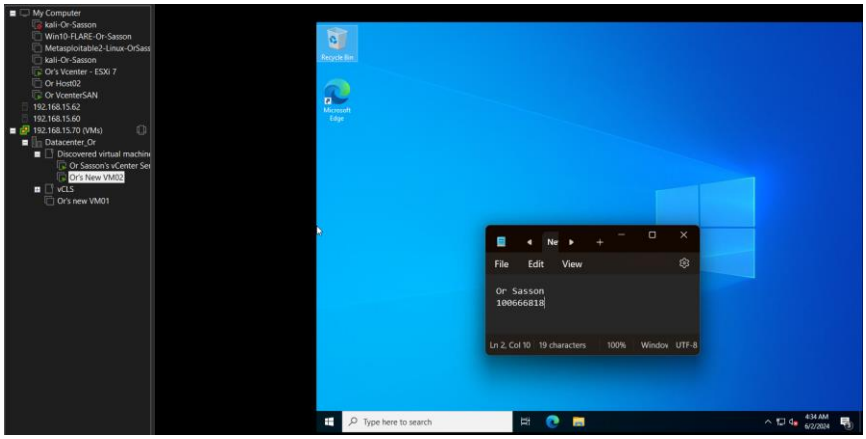

New Backup Repository
Summary
You can copy the configuration information below for future reference.
Name
Server
Repository
Mount Server
Review
Apply
Summary
Summary:
Windows backup repository 'OrBackup' was successfully saved.
Mount host: OrVcenterSAN
Backup folder: F:\OrBackup
Write throughput: unlimited
Max parallel tasks: 4
< Previous
Next >
Finish
Cancel


New VMware Server
Summary
You can copy the configuration information below for future reference.
Name
Credentials
Apply
Summary
Summary:
VMware ESXi server '192.168.15.60' was successfully saved.
Host info: VMware ESXi 7.0.3 build-19482537
Connection options:
User: BackupUser
Port: 443
File
Edit
View
Or Sasson
100666818
Ln 2, Col 10
19 characters
100%
Windov
UTF-8

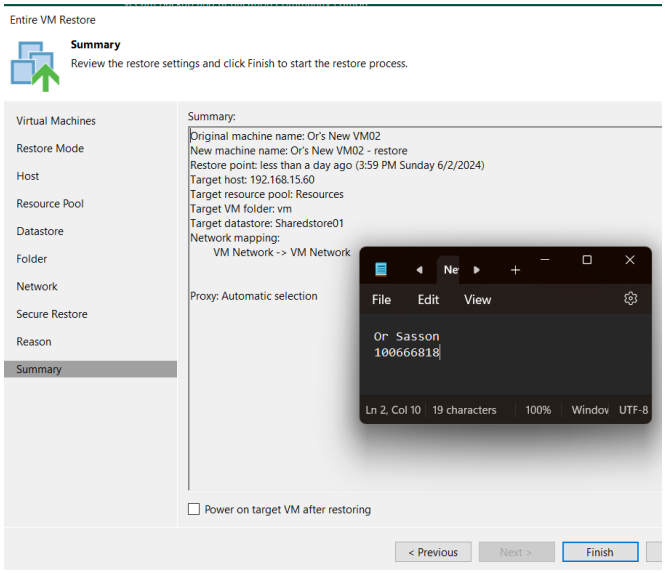

New Backup Job
Summary
You have successfully created the new backup job.
Name
Virtual Machines
Storage
Guest Processing
Schedule
Summary
Summary:
Configuration has been successfully saved.
Name: OrBackup VM02
Target Path: F:\OrBackup
Type: VMware Backup
Source items:
Or's New VM02 (192.168.15.60)
PowerShell cmdlet for starting the job:
Get-VBRJob -Name "OrBackup VM02" | Start-VBRJob
Run the job when I click Finish
< Previous
Next >
Finish
Cancel

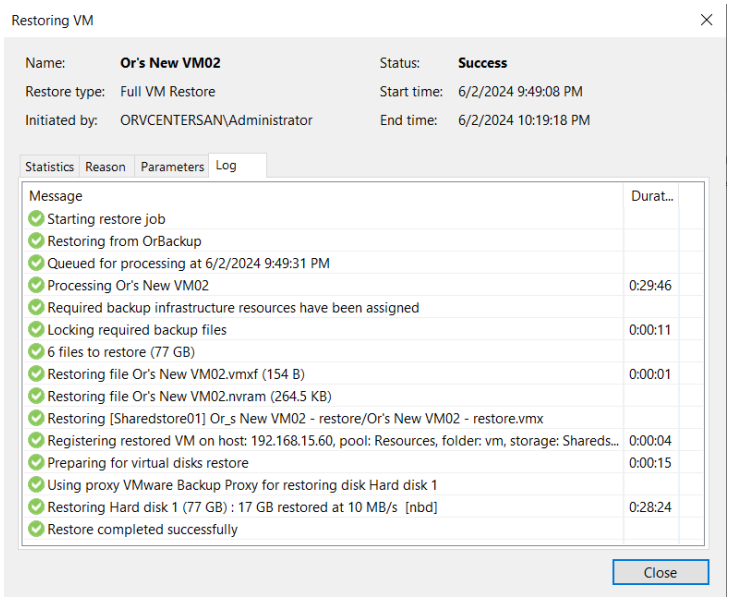

Veeam Backup and Replication Community Edition
Home
View
Job
Start
Stop
Retry
Active Full
Statistics
Report
Edit
Clone
Disable
Delete
Job Control
Details
Manage Job
Jobs
Backup
Backups
Disk
Last 24 Hours
Success
All jobs
Name
Type
Obje...
Status
Last Run
Last Result
Next Run
Target
OrBackup VM02
VMware Backup
1
Stopped
3 hours ago
Success
6/2/2024 10:00 PM
SUMMARY
Duration:
23:03
Processing rate:
15 MB/s
Bottleneck:
Source
DATA
Processed:
21.2 GB (100%)
Read:
17.1 GB
Transferred:
9.8 GB (1.7x)
STATUS
Success:
1
Warnings:
0
Errors:
0
THROUGHPUT (ALL
Speed: 20 MB/s
Home
Inventory
Backup Infrastructure
Storage Infrastructure
Tape Infrastructure
Files
Name
Status
Action
Duration
Or's New VM02
Succ...
Processing Or's New VM02
22:26
Load: Source 99% > Proxy 79% > Network 14% > Target 0%
Primary bottleneck: Source
Job finished at 6/2/2024 4:19:29 PM
1 job selected
Connected to: localhost
Build: 12.1.1.56
Community Edition

After performing a successful backup of the VM02, I deleted the files I created before on the VM so I can restore them again from the backup.



Then, I restored it with Veeam backup software.

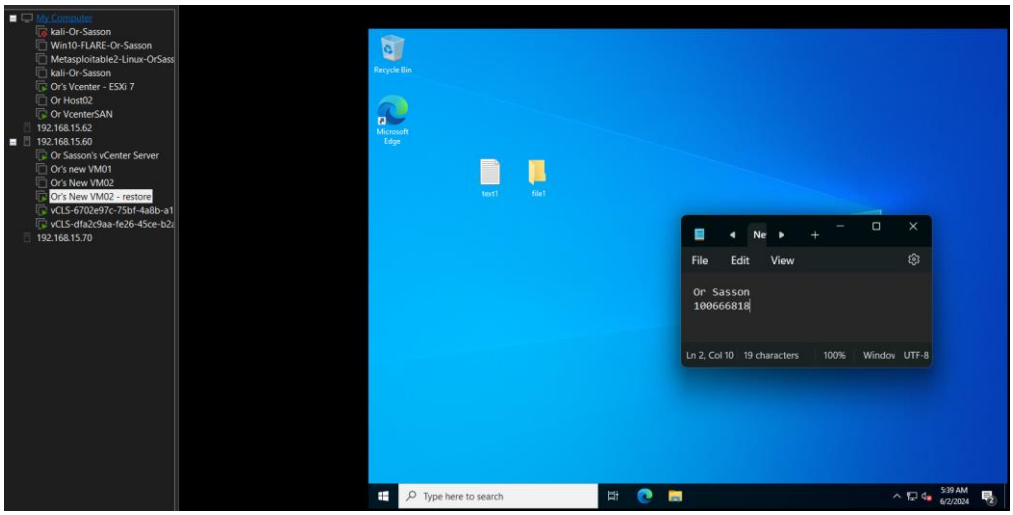




Virtual machine	Status	Used space	Guest OS	Host name	Host CPU	Host memory
Or Sasson's vCenter Server	Normal	53.82 GB	Other 3.x or later Linux (64-bit)	localhost	1.1 GHz	7.79 GB
Or's new VM01	Normal	385.44 KB	Microsoft Windows Server 2...	Unknown	0 MHz	0 MB
vCLS-dfa2c9aa-fe26-45ce-b2a2-9fd6c1dc3cd6	Normal	459.24 MB	Other 3.x or later Linux (64-bit)	photon3-hdcs	51 MHz	126 MB
vCLS-6702e97c-75bf-4a8b-a1d9-b4204931a03d	Normal	459.37 MB	Other 3.x or later Linux (64-bit)	photon3-hdcs	63 MHz	125 MB
Or's New VM02	Normal	22.24 GB	Microsoft Windows Server 2...	Unknown	0 MHz	0 MB
Or's New VM02 - restore	Normal	16.98 GB	Microsoft Windows Server 2...	Unknown	0 MHz	0 MB

Quick filters... 6 items

Now you can see the files on the restored VM which I deleted before from the desktop.



Run the test plans to ensure the outcomes match with the test plans.

Test plan 1

Test Type	Description	Expected results	Actual results	Status	Comments\ Issues
Internal ping	Connectivity between all VMs and to the hosts. Also, connectivity check between the hosts.	Successful ping responses from all VMs and hosts	Responded correctly	Successful	No issues.
External ping to 8.8.8.8 (Google)	Connectivity between all VMs and hosts to the internet.	Successful ping responses to the external network	Responded correctly	Successful	No issues.

Test plan 2

Test Type	Description	Expected results	Actual results	Status	Comments\ Issues
High Availability	Failover test while shutting down one of the hosts with the VMs	Successful migration of all VMs from the failed host to an alternative one.	The VMs moved successfully to the other host	Successful	No issues.
Live Migration	Performing live migration of one of the VMs while it is powered on	Successful migration while the VM is in the same state	The VM migrated well and operated as normally	Successful	No issues.
Snapshot	Create a snapshot of one of the VMs	Successful creation	Created the snapshot successfully	Successful	No issues.
Backup	Performing a test backup with Veeam on VM02 and then restore it.	Successful backup and restore of VM02	Successful backup and restore	Successful	No issues.

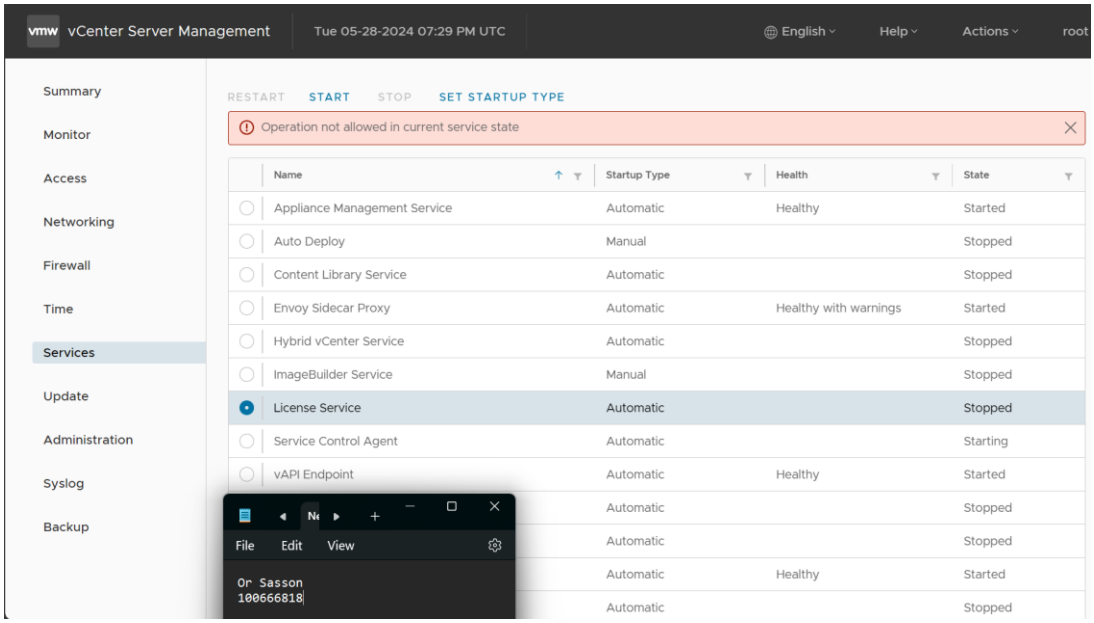
Implement, test, and document the protection plan

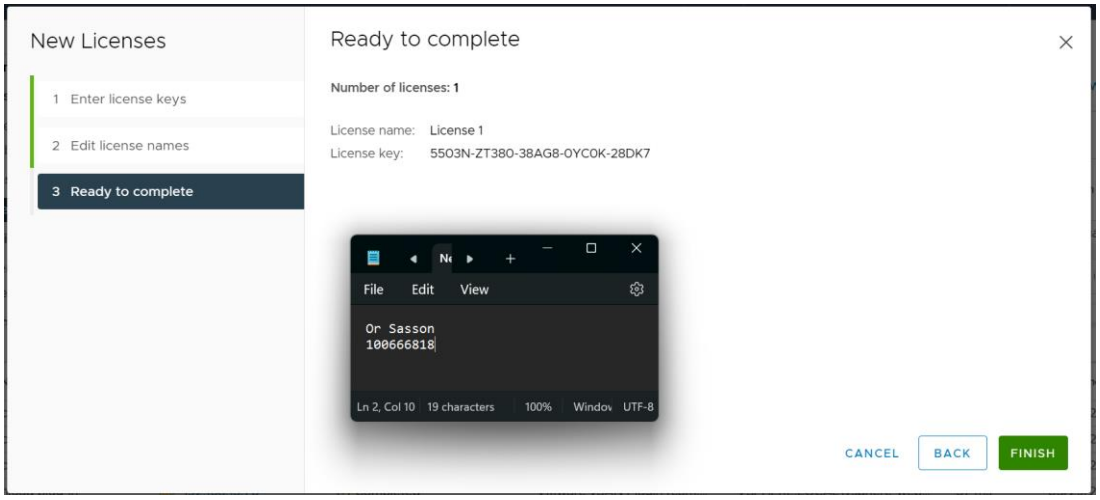
Before and during the implementation, I identified the critical assets of the business and I ensured that the server room got the right access control including keycard access and security cameras along with cooling, humidity control and fire suppression systems. Everything been tested and works well. During the implementation in part B, the firewall OVF file was deployed successfully along with high availability, antivirus software and backup solution configured to ensure the integrity, confidentiality and availability of the infrastructure. Also, the VMs and servers got patched and they are up to date with update plan in place.
Security controls with strong encryption and password been implemented as well along with the education of the employees to ensure the systems are as secured as possible.

It was recommended that the business review logs regularly and deploy IDS/IPS systems along with vulnerability scanning with OpenVAS, penetration testing to simulate attacks with Metasploit for instance and having an incident response plan in place to monitor, detect, response and recover from incidents quickly.

Analyse errors and troubleshoot if you have any issues.

1. I couldn't assign the license for the vSphere and after troubleshooting I realized that the License Service stopped so I started this service again and solved the issue.





2. HA worked for me only on host01 out of 3 hosts, after troubleshooting I realized that I should shut down host03 and give more resources to host02. After giving more resources to the host02, the HA performed well between host01 and host02. My computer has only 16GB of RAM so I could not perform the HA on all 3 hosts together.

Limit access via enforcing strict access security policies

Securing the systems could be done by the following steps:

- Implementation of Role-based access control which define the roles in the business such as administrator or dentist for instance and assign the permission based on the role and not based on an individual user.
- Give the least privilege so the user can access to the minimum level of access which necessary for his tasks.
- Using multi-factor authentication (MFA) to identify user. This can be done by receiving SMS with a verification code in addition to the username and password for instance.
- Having password policies in place to ensure complex passwords for all users and define how often they need to be changed.
- Monitoring access logs and permissions to detect any unauthorized access attempts and investigate it.
- Have network segmentation in place including VLANs and firewalls.
- Monitor the system, follow patterns, detect and response on time to security incidents.
- Have training session and awareness to educate employees about best practices and security.

User documentation

In this user documentation I will provide a guidance for both users and administrators of Clean Dental's newly virtual infrastructure which will ensure the smooth operation and maintenance of the servers and networks.

There are 3 ESXi hosts which contains:

- vCenter
- Email server using iRedMail
- Microsoft terminal server
- best practice server

In addition, there is a SAN server on Windows server 2022 including Veeam backup solution.

OVF file with firewall was deployed to protect the network.

IP addresses:

Host01: 192.168.15.60
Host02: 192.168.15.61
Host03: 192.168.15.62
SAN Server: 192.168.15.80
vSphere: 192.168.15.70
VM01: 192.168.15.65
VM02: 192.168.15.66
VM03: 192.168.15.67

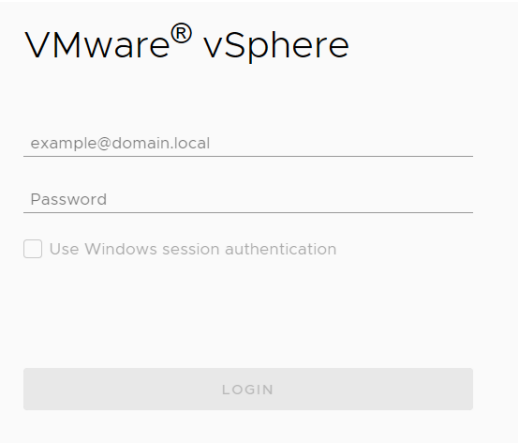
All hosts can be managed through the web by typing https://192.168.15.60 for instance. After typing the address, you will be required to log in with the root user and your password.
It is possible to manage the VMs on the hosts from the web as well.

The login page:



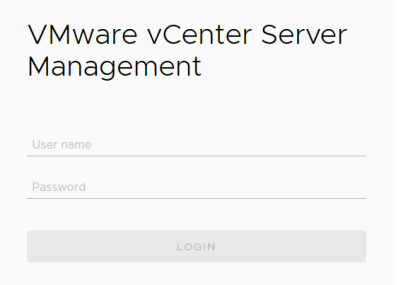
All hosts are linked to vSphere which gives you the ability to manage the virtual servers, VMs, high availability along with monitoring the performance through the web from the following address: https://192.168.15.70:443.
After the login you will need to access through the user: administrator@vsphere.local with your password.

Login page of vSphere:



The vCenter and the vSphere can be managed from the address: https://192.168.15.70:5480.
The access can be gained with the root user and the password.

Login page of VMware vCenter Server Management:



The SAN server can be managed from VMware Workstation, and you can manage it from the administrator user. On The SAN server you can manage and control the Veeam backup solution.

All VMs can be managed from the administrator user and can be accessed by permitted users depending on their privileges. You can access them via RDS.

All systems are configured to ensure automatic patches and updates during nighttime, and the Veeam have a schedule for backup tasks on all systems.

If any issue raised, you are welcome to contact the IT helpdesk via email or their phone number in their operational hours.

References

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Cybersecurity Best Practices | Cybersecurity and Infrastructure Security Agency CISA

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Troubleshooting with ESXi Shell – esxsi.com

VMware vSphere Documentation